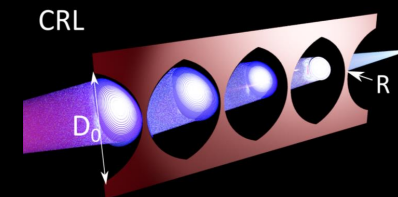
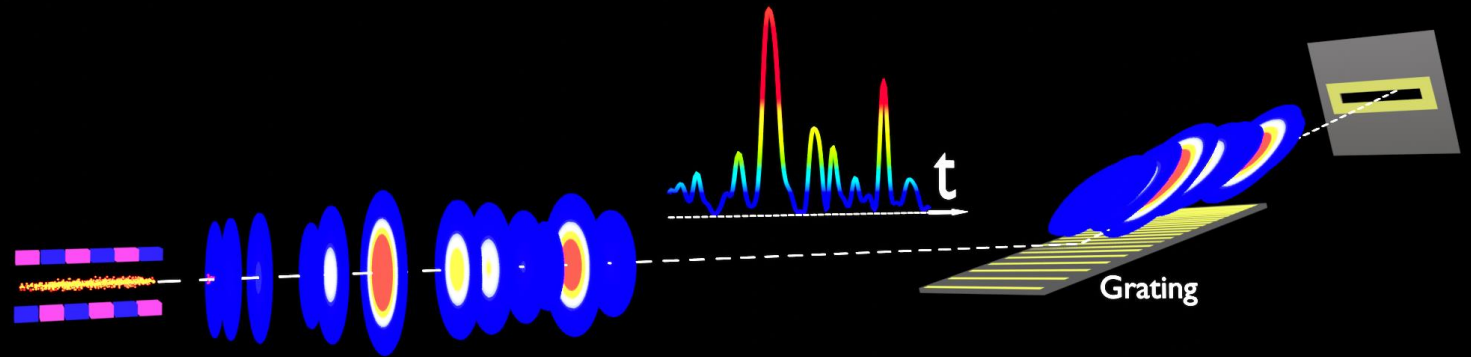
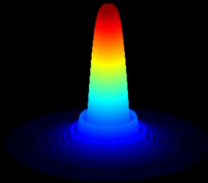
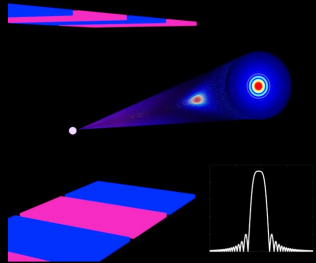
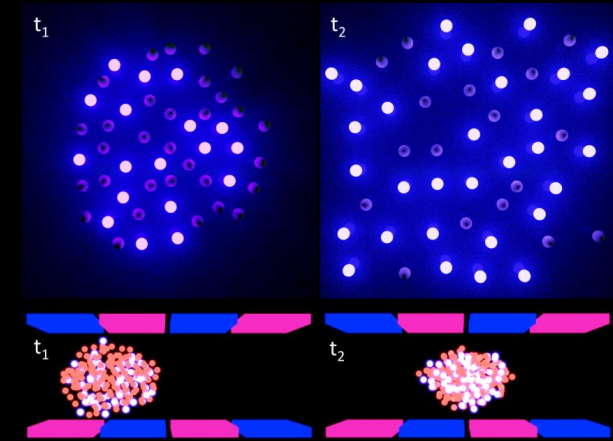
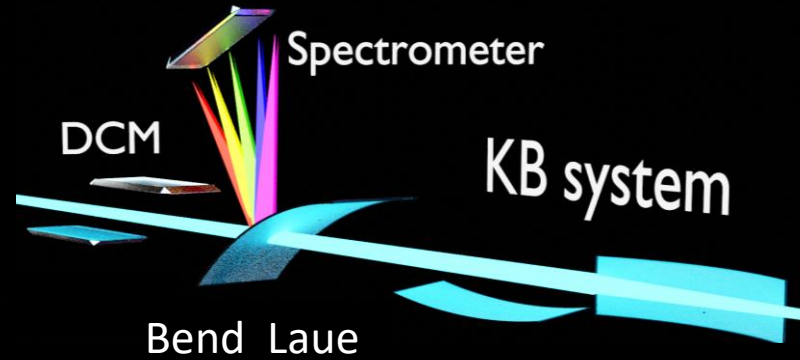
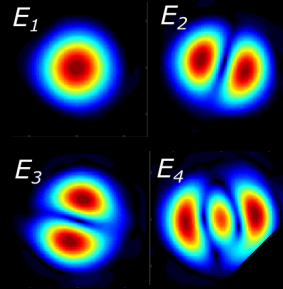
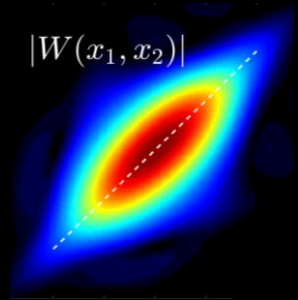


Basics of Coherence



Ruslan Khubbutdinov

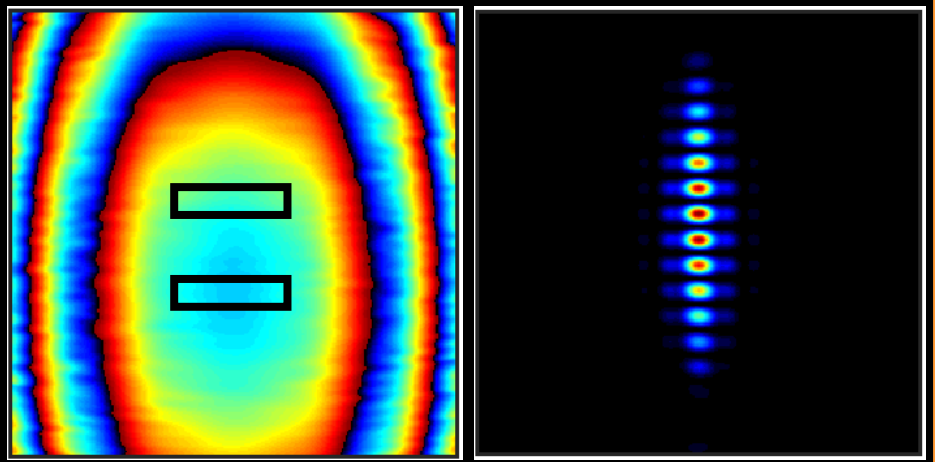
22.05.25, Hamburg, DESY

Beamline Optics Design and Simulation Workshop 2025

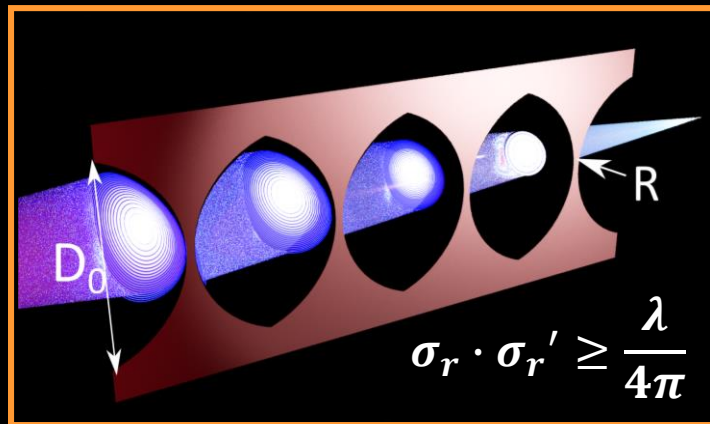
Deutsches Elektronen-Synchrotron DESY, Notkestraße 85, 22607 Hamburg, Germany



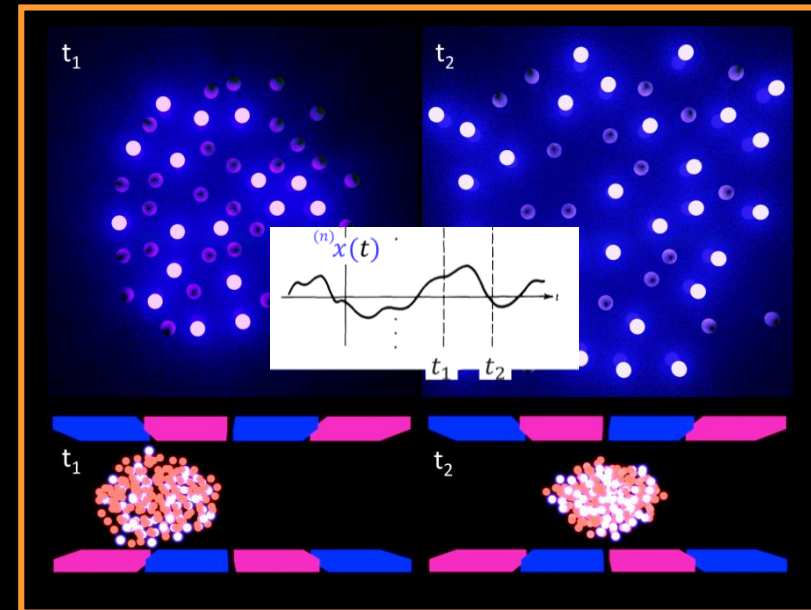
Phase correlation ?



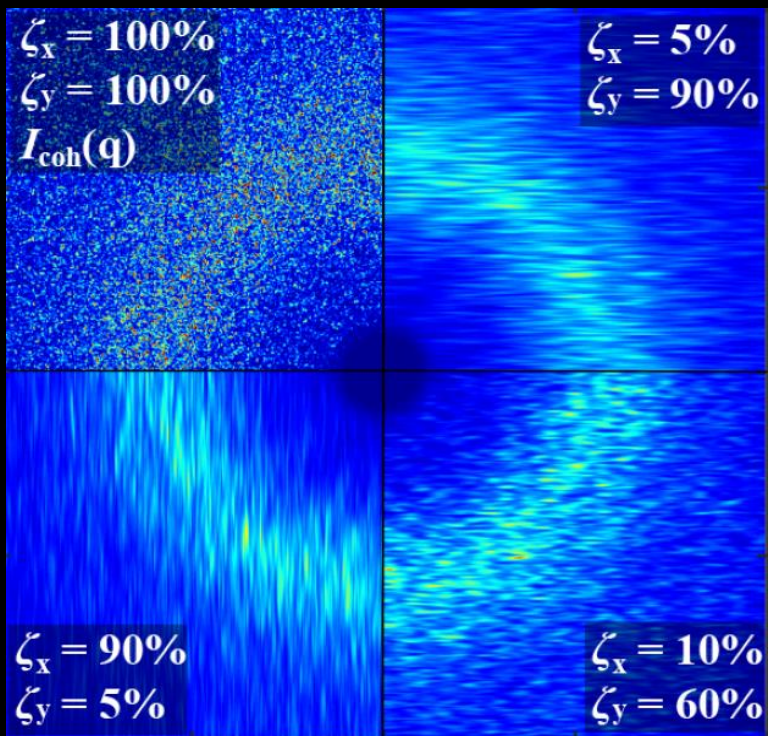
Focusing ability/property?



Statistical property?

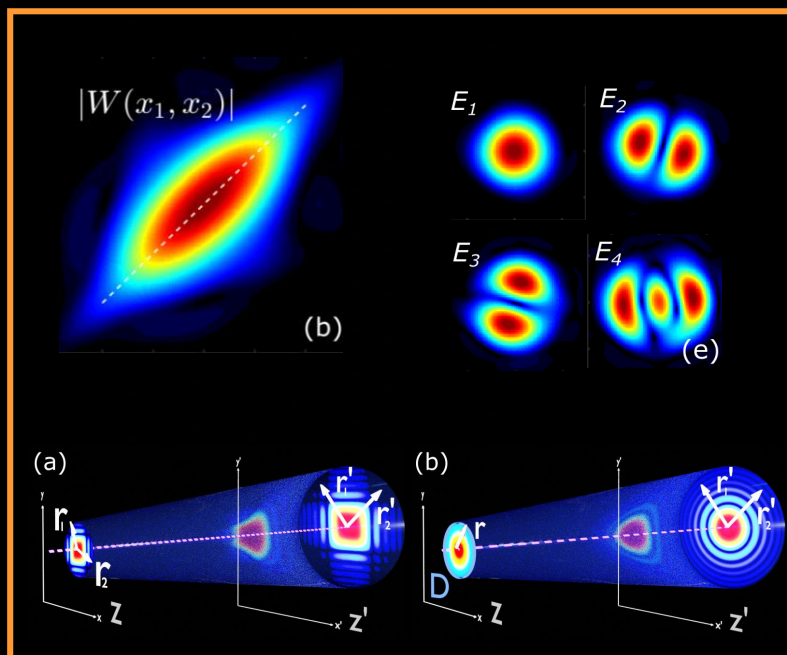


Contrast of the diffraction patterns?

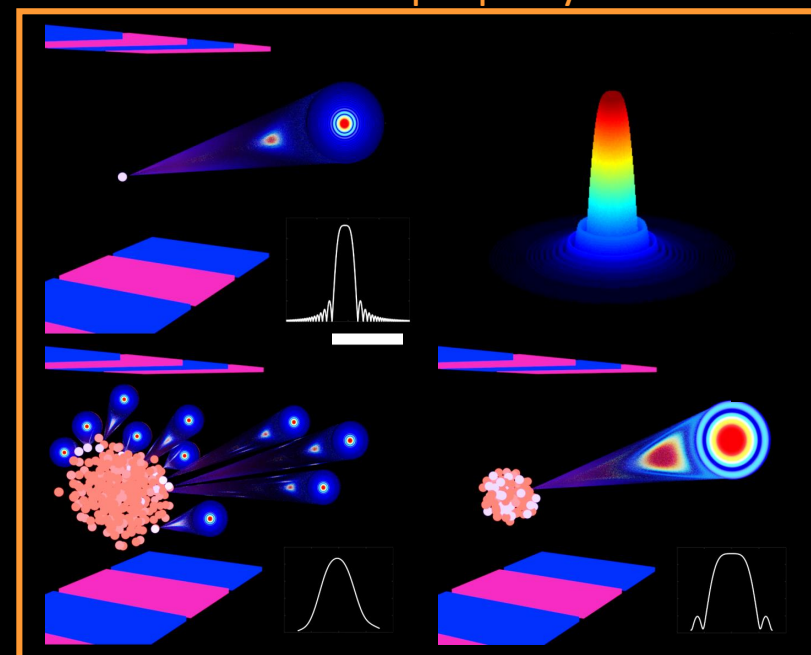


Coherence ?

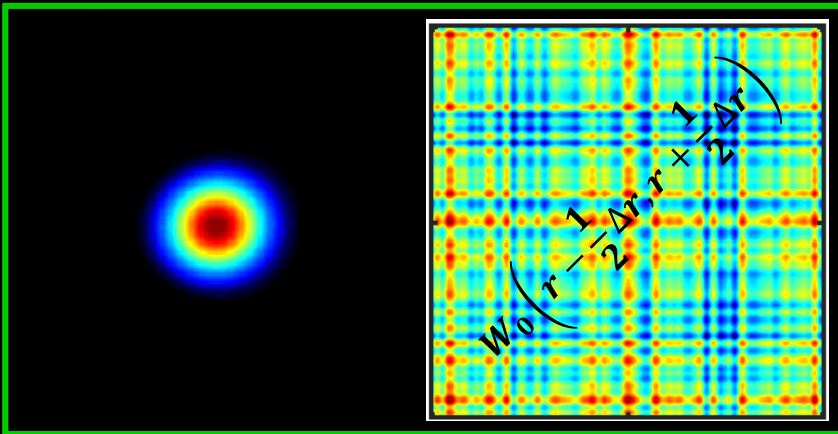
Photon beam property?



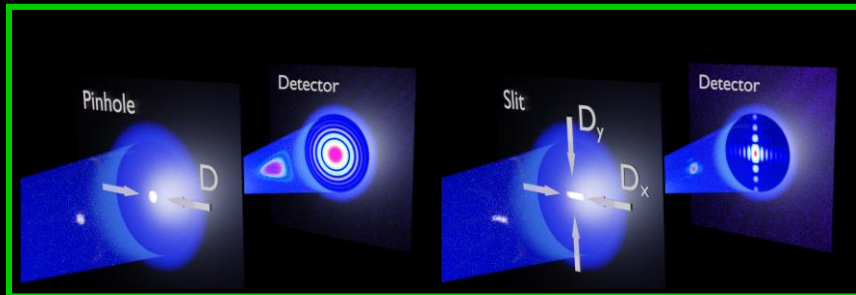
Source property?



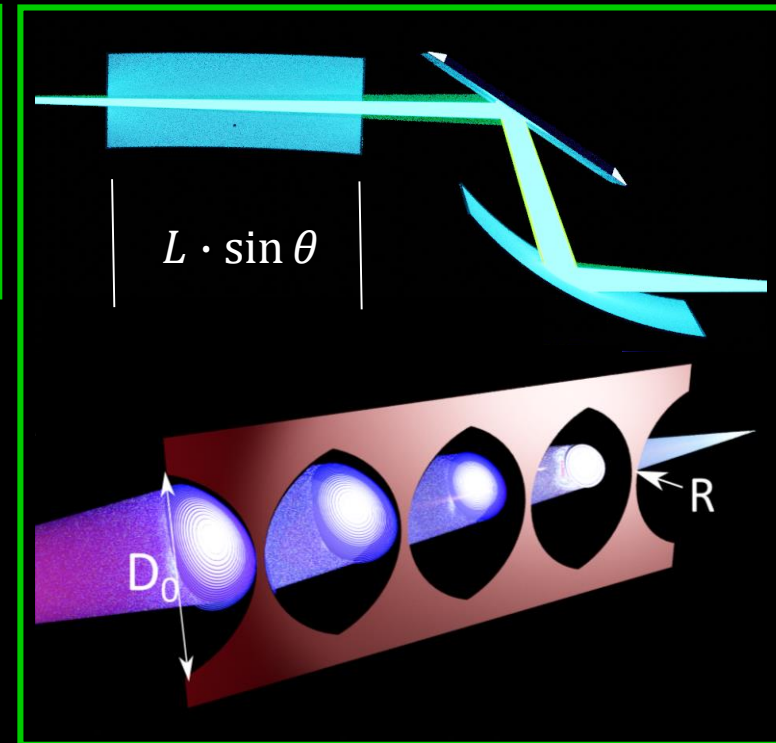
Vibrations ?



BDS?

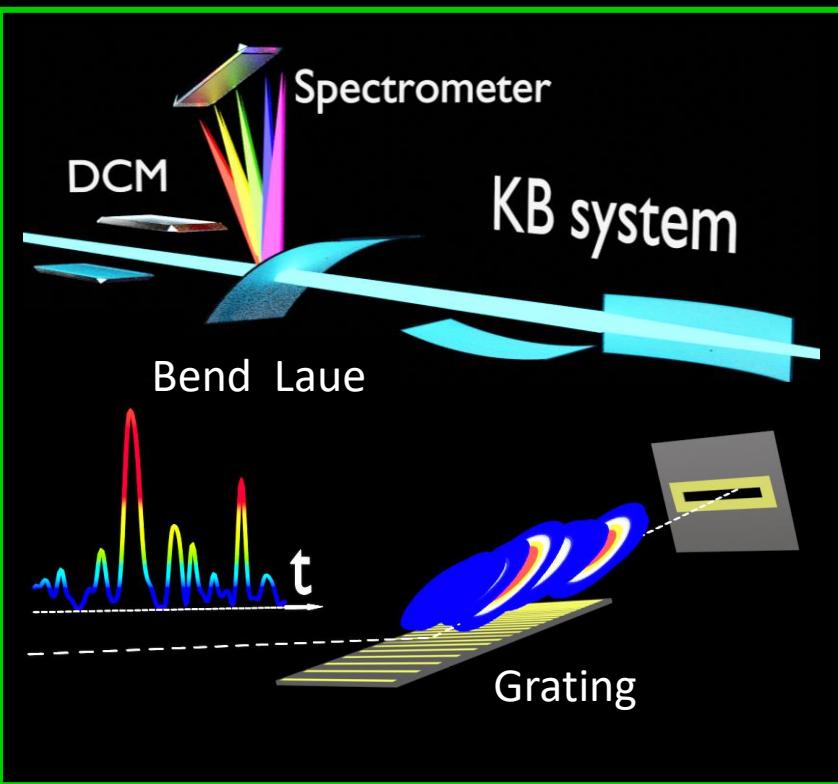


Aperture?

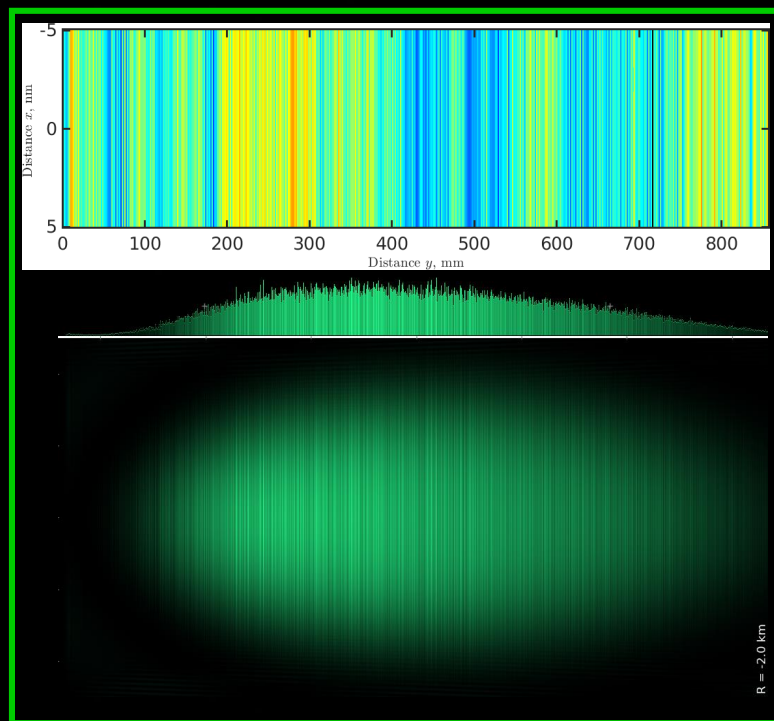


Effects on Coherence ?

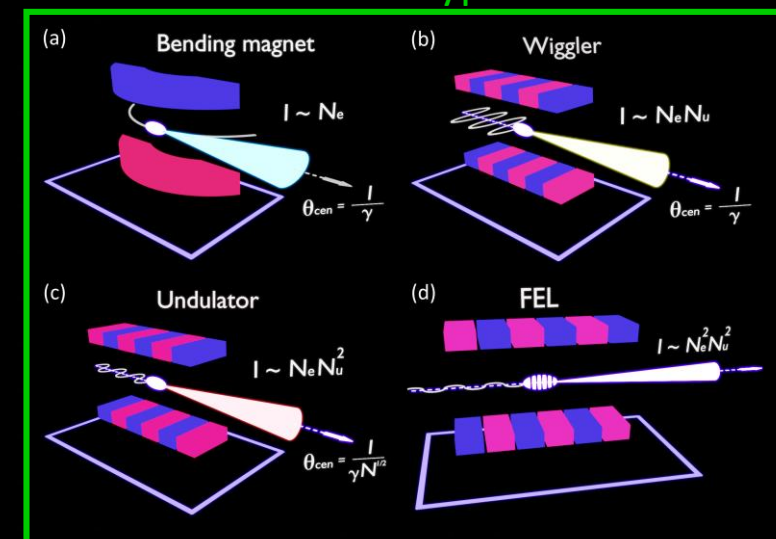
Optics type?



Surface?

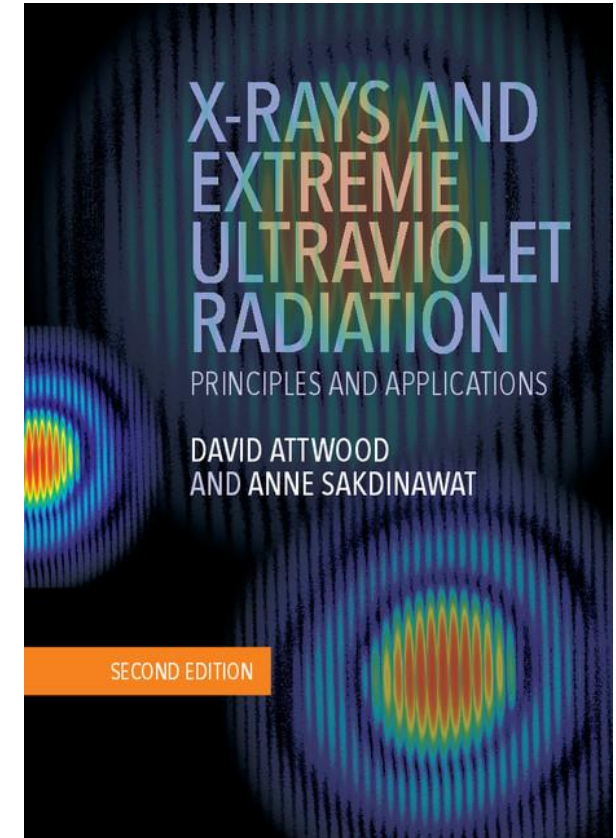
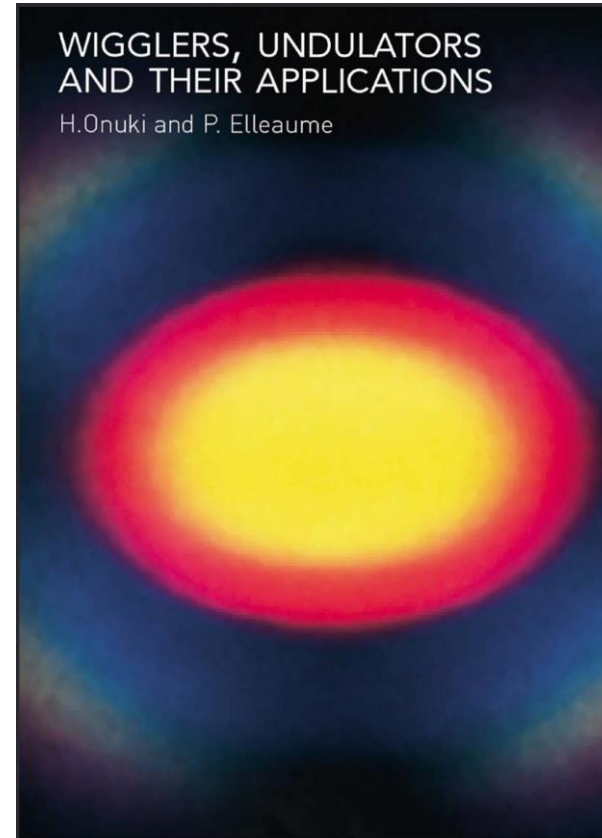
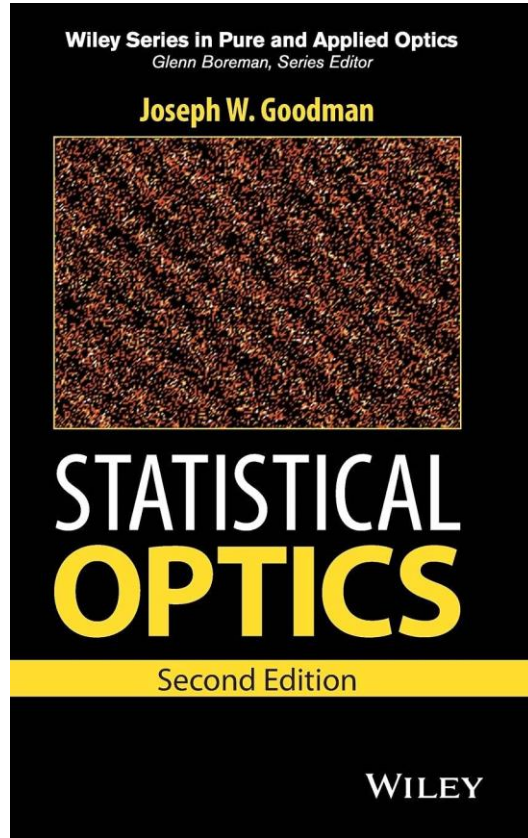
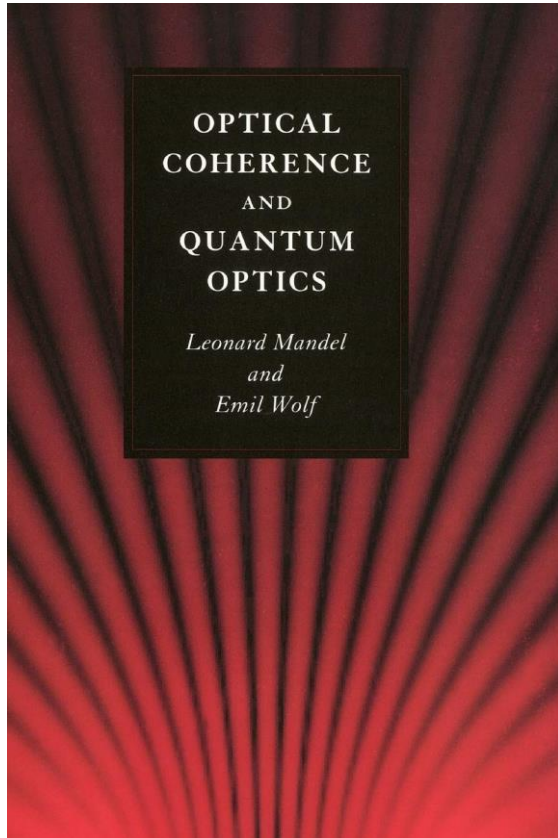


Source type?



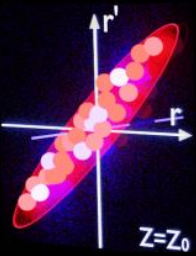
Literature

Pillars of coherence and SR



* adapted pictures

Coherence



“Coherence is the manifestation of the purity of a quantum state of light or matter”

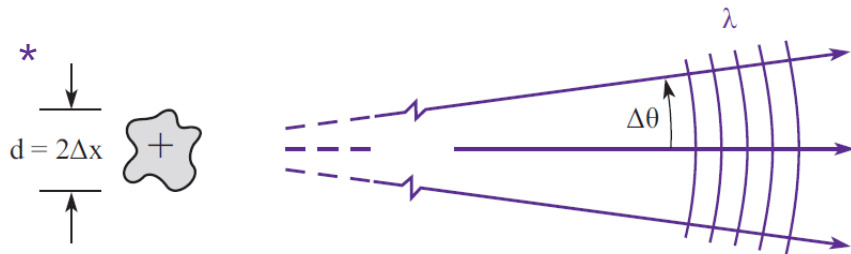
Notions time and space coherence are closely related to Heisenberg's Uncertainty Principle, which define limits of quantum mechanical distributions (Gaussian):

Space

space - momentum

$$\Delta x \cdot \Delta k > 1/2$$

spatial coherence is related to the finite source size and the characteristic emission angle of the radiation



The smallest source size (point source) d resolvable with finite wavelength λ can be determined

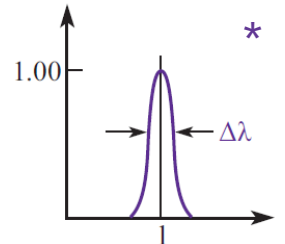
Time

time - frequency

$$\Delta \mathcal{E} \cdot \Delta \tau \geq \hbar/2$$

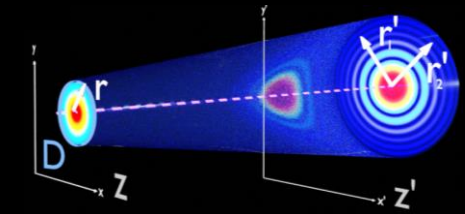
$\Delta \mathcal{E}$ is the rms bandwidth of the photon energy required to generate a pulse of rms duration $\Delta \tau$

$$\Delta \omega \cdot \Delta \tau \geq 1/2$$



The larger the range of frequencies $\Delta \omega$ a wave contains, the faster the wave decorrelates (Coherence time is small).

Coherence



“Coherence is the manifestation of the purity of a quantum state of light or matter”

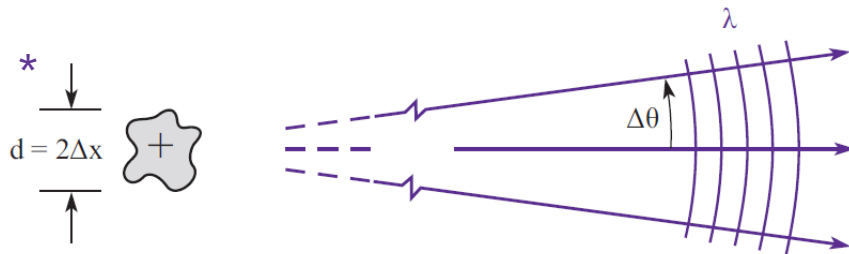
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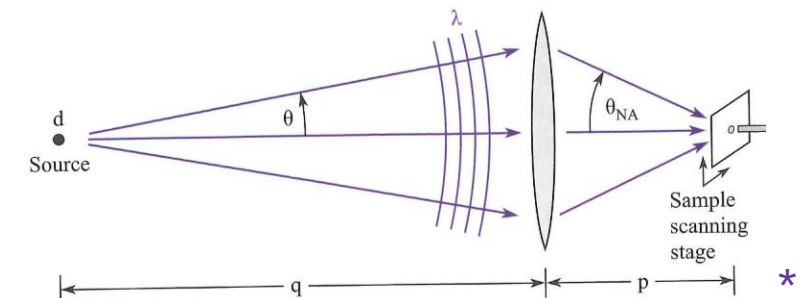


Coherent phase-space area (Emittance)

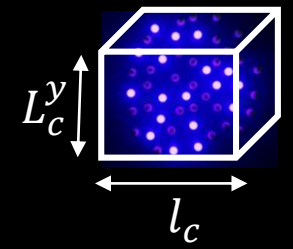
$$\varepsilon_{coh} = \Delta x \cdot \Delta \theta \geq \lambda/4\pi$$

Diffraction limit

$$d = \frac{2\pi}{\lambda\theta} \approx \frac{2\pi}{\lambda\sin(\theta)} \approx \frac{2\pi}{\lambda \cdot NA}$$



Coherence and Coherence length



“Coherence is the manifestation of the purity of a quantum state of light or matter”

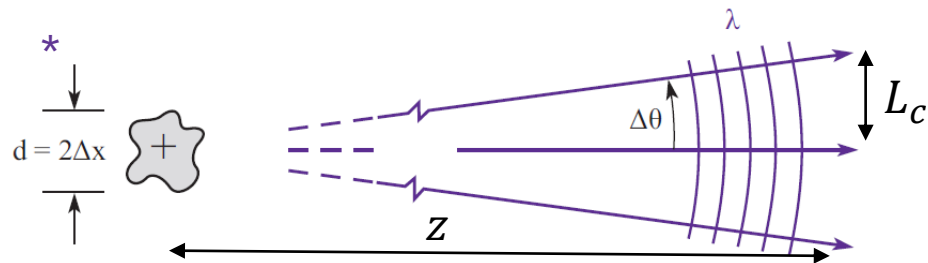


Space

space - momentum

$$\Delta x \cdot \Delta \theta \geq \lambda / 4\pi$$

spatial coherence is related to the finite source size and the characteristic emission angle of the radiation



Transverse coherence length

(determined by the source size)

$$L_c \approx z \Delta \theta \approx \frac{\lambda z}{2\pi d}$$

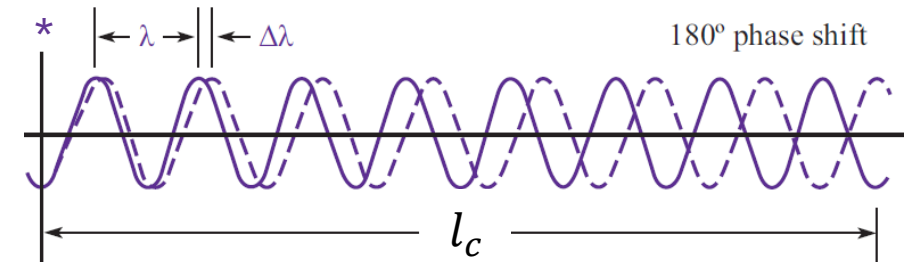


Time

time - frequency

$$\Delta \omega \cdot \Delta \tau \geq 1/2$$

The larger the range of frequencies $\Delta \omega$ a wave contains, the faster the wave decorrelates



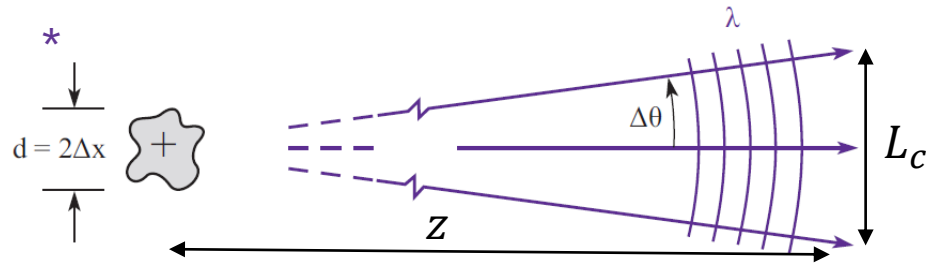
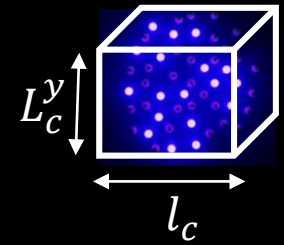
Longitudinal coherence length

(determined by monochromaticity)

$$l_c \approx \frac{\Delta \lambda}{\lambda^2}$$

Coherence

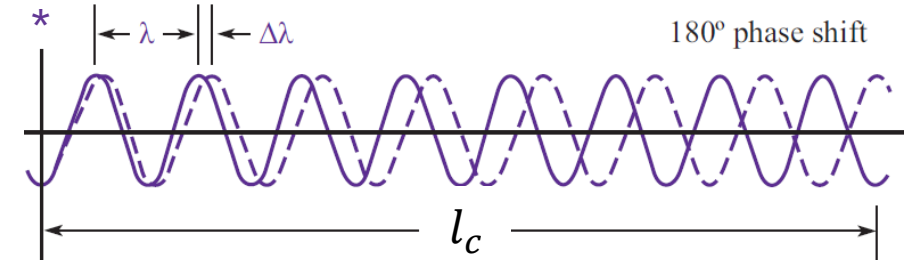
“Coherence is the manifestation of the purity of a quantum state of light or matter”



Transverse coherence length
(determined by the source size)

$$L_c \approx z \Delta \theta \approx \frac{\lambda z}{2\pi d}$$

Theorem?



Longitudinal coherence length
(determined by monochromaticity)

$$l_c \approx \frac{\Delta \lambda}{\lambda^2}$$

For a Gaussian distributed source with the RMS size of Σ :

$$L_c = \frac{\lambda z}{2\pi \Sigma} \approx 0.16 \frac{\lambda z}{\Sigma}$$

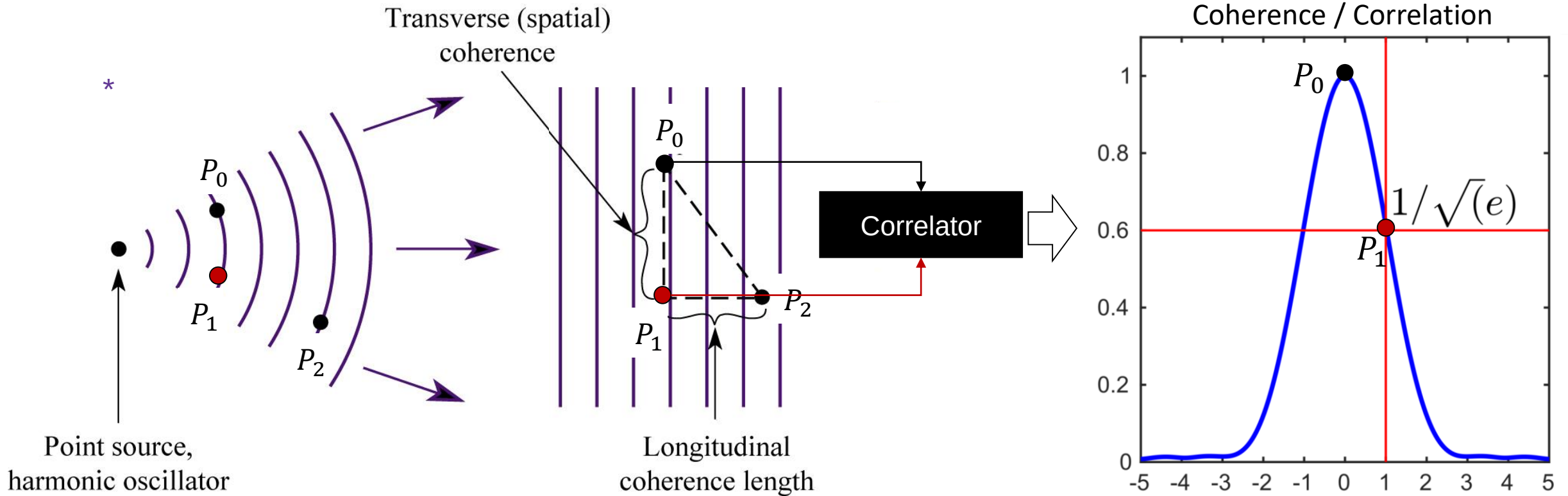
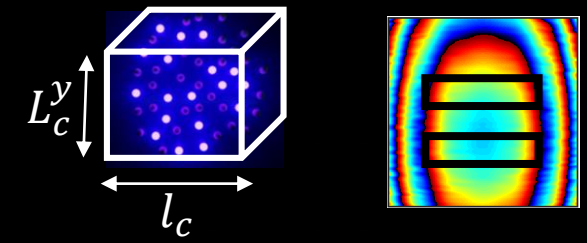
For a flat rectangular beam with size Δ : $L_c = \frac{\lambda z}{2\Delta}$

For $\lambda/\Delta\lambda = 7000$ (Si 111), $\lambda = 0.1$ nm, $l_c = 0.7$ μ m

$l_c \gg \text{max path difference}$ to ensure good contrast

Coherence

“Coherence is the manifestation of spatial-temporal correlations”



The concept of coherence has long been associated with interference since it is the most direct way of measuring it

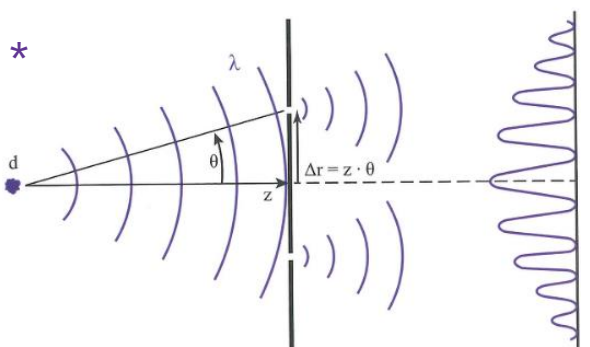
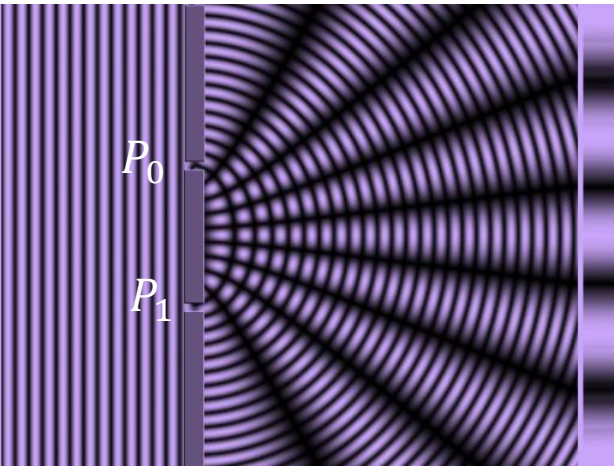
Coherence

“Coherence is the manifestation of spatial-temporal correlations”

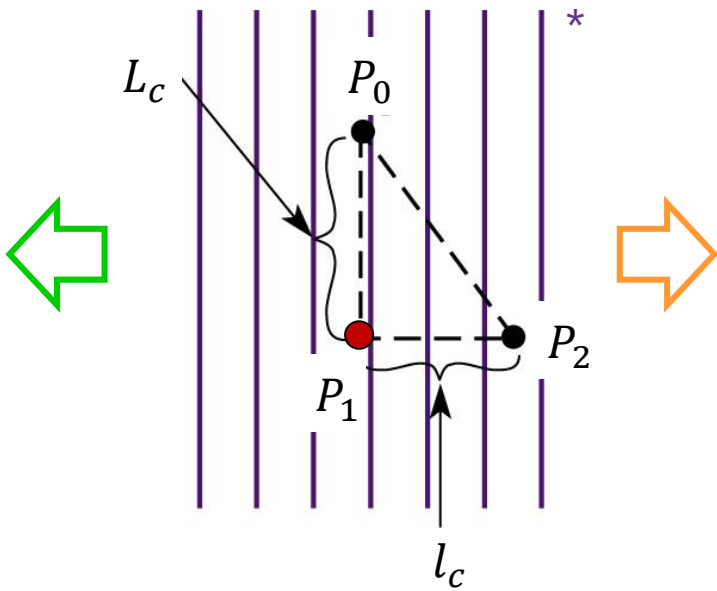
The concept of coherence has long been associated with interference since it is the most direct way of measuring it

Spatial correlation

Young’s double slit experiment
Spatial Coherence length



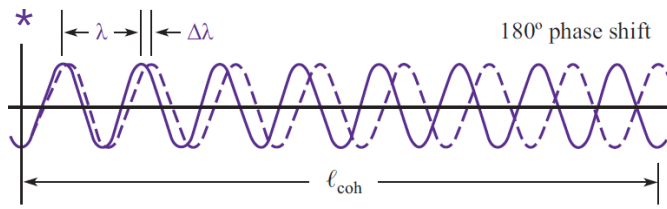
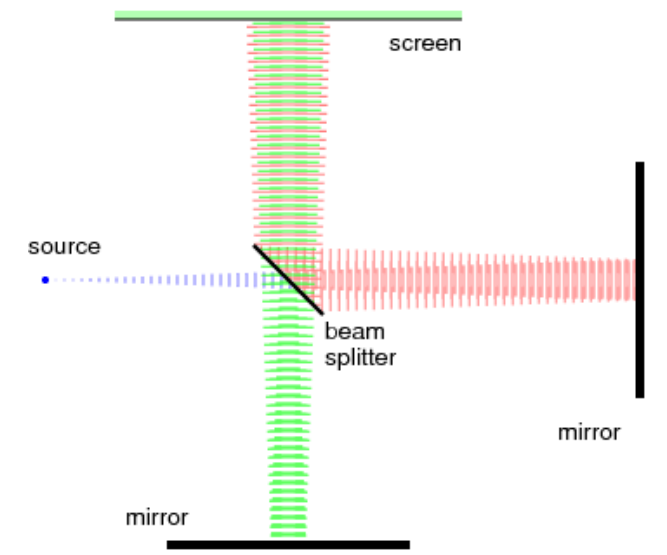
Ability of a light beam to form fringes with a delayed version of itself



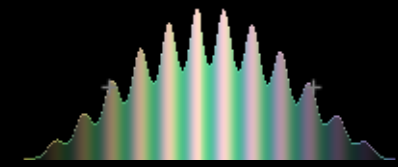
Ability of spatially separated points in a wavefront to form fringes.

Temporal correlation

Michelson interferometer
Longitudinal Coherence length



Coherence

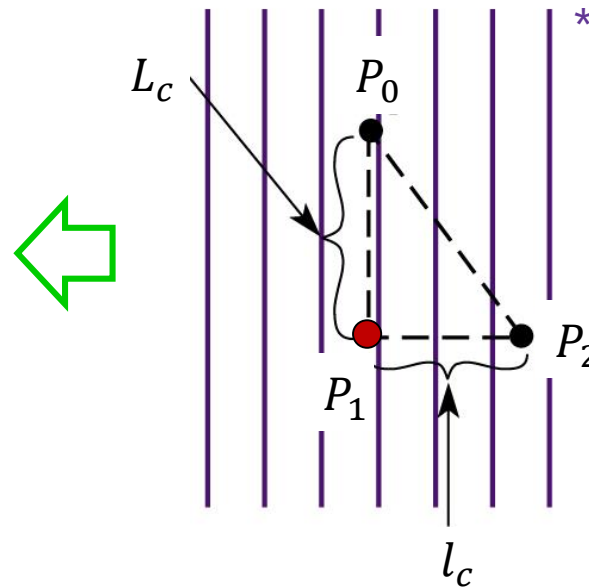
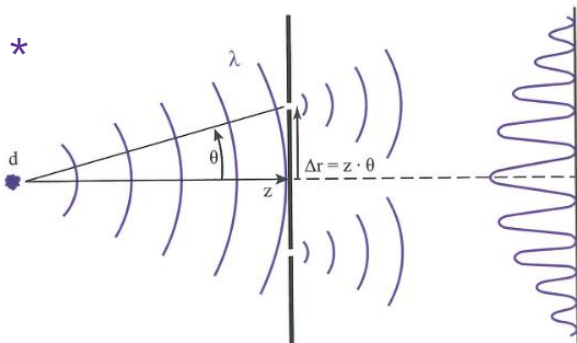
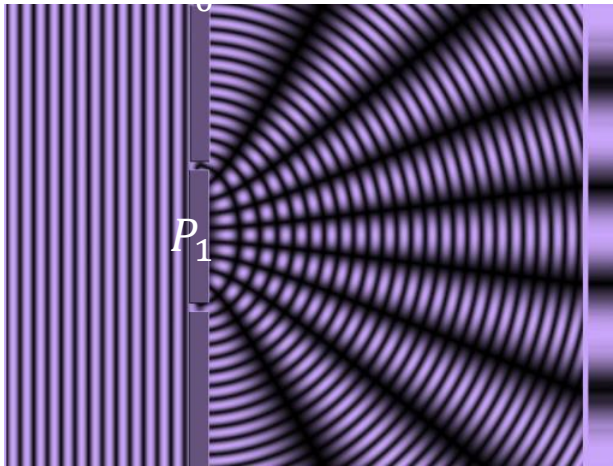


“Coherence is the manifestation of spatial-temporal correlations”

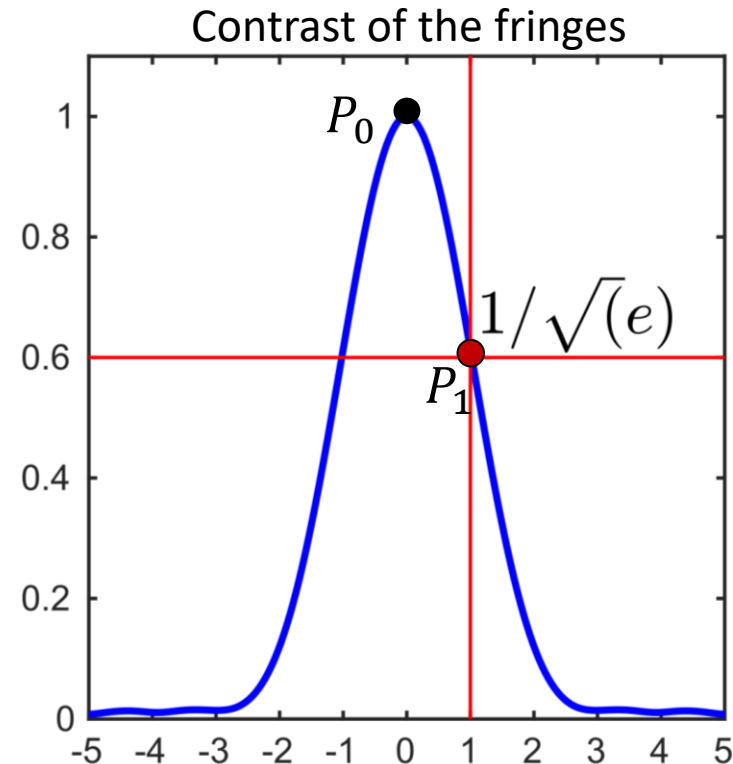
The concept of coherence has long been associated with interference since it is the most direct way of measuring it

Spatial correlation

Young's double slit experiment Spatial Coherence length



Ability of spatially separated points in a wavefront to form fringes.

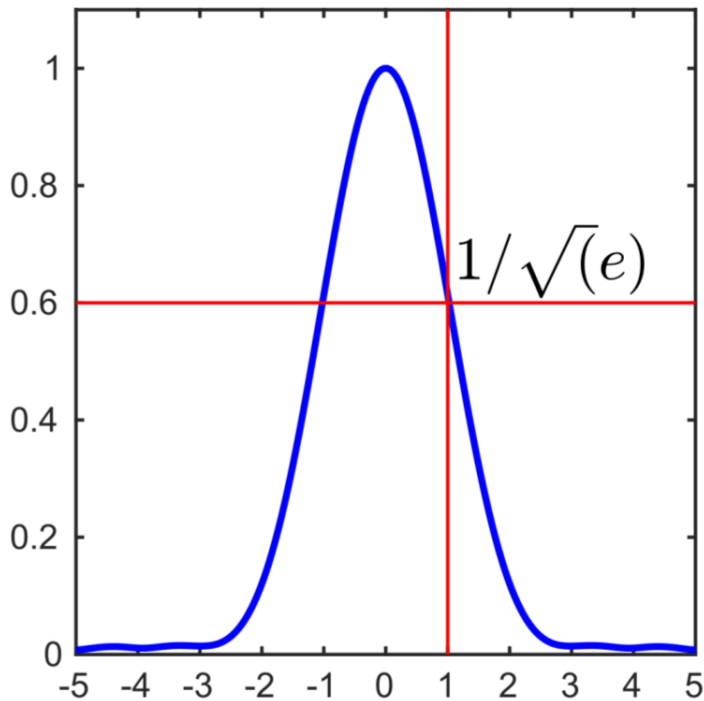
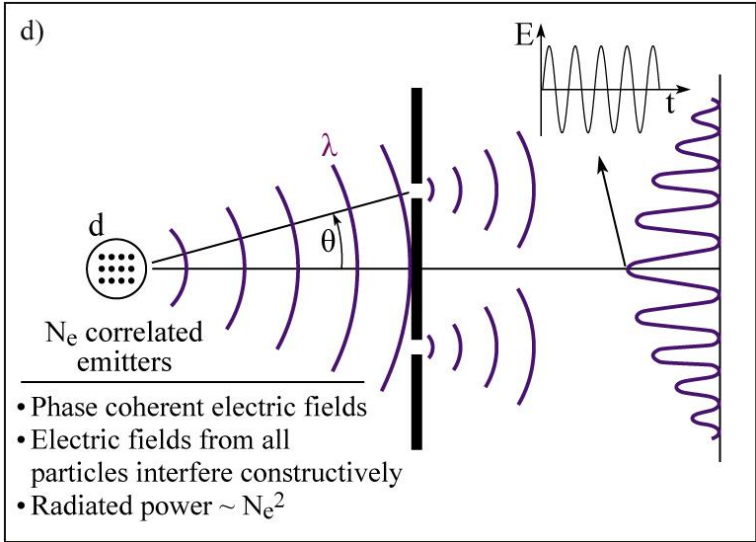
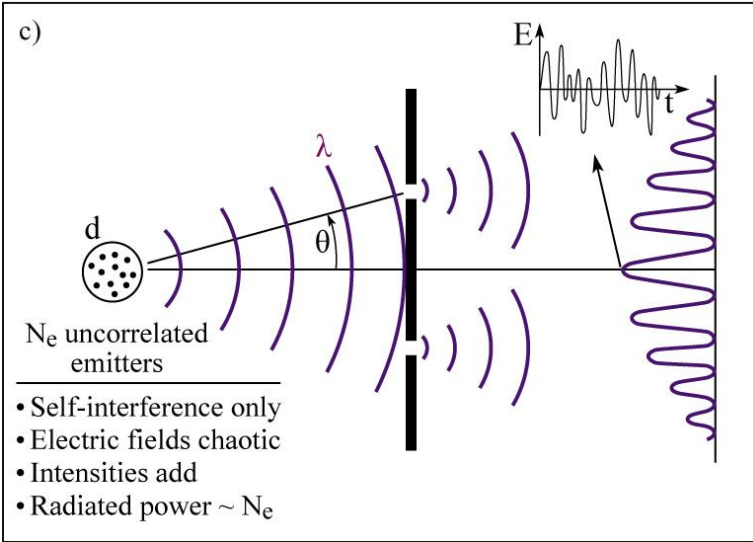
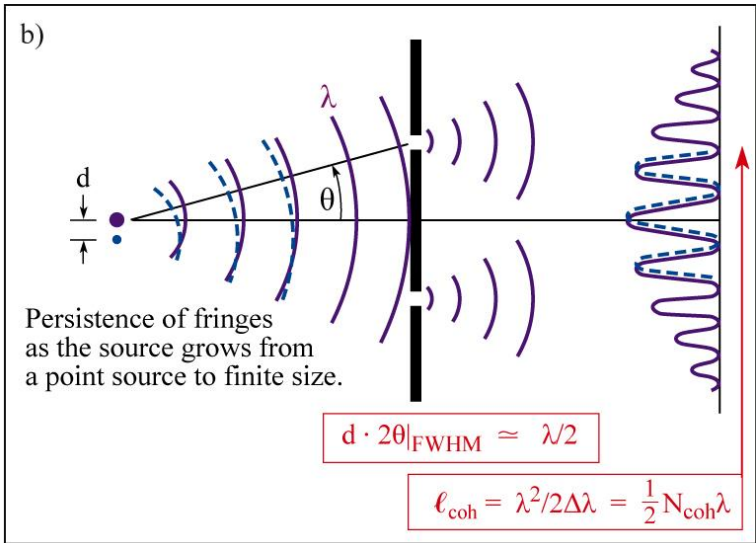
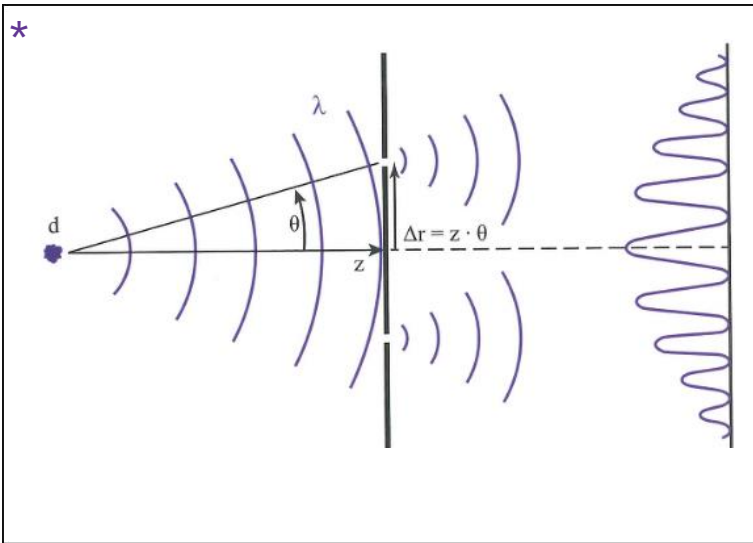
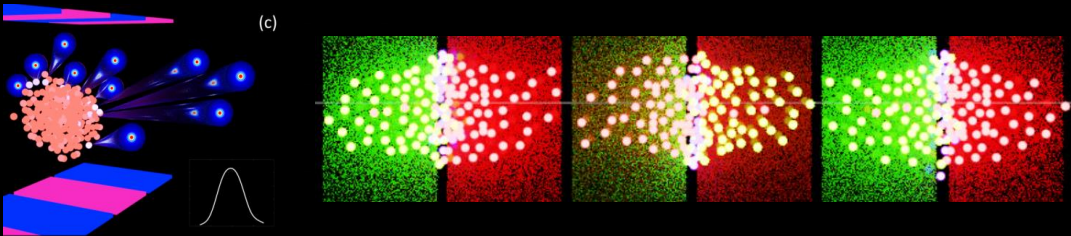


Quantitatively, the transverse coherence length L_c is described as the distance between the two narrow slits in the Young's experiment which drops the interference fringes contrast to $\exp(-1/2) = 0.6$.

$$I(\mathbf{q}) = I_1 + I_2 + 2\sqrt{I_1 I_2} |\mu_{12}| \cos(\mathbf{q} * \Delta \mathbf{x})$$

Coherence

“Coherence is the manifestation of spatial-temporal correlations”



Quantitatively, the transverse coherence length L_c is described as the distance between the two narrow slits in the Young’s experiment which drops the interference fringes contrast to $\exp(-1/2) = 0.6$.

Coherence – Statistical property!

Mutual Coherence Function (MCF)

Space-time (MCF)

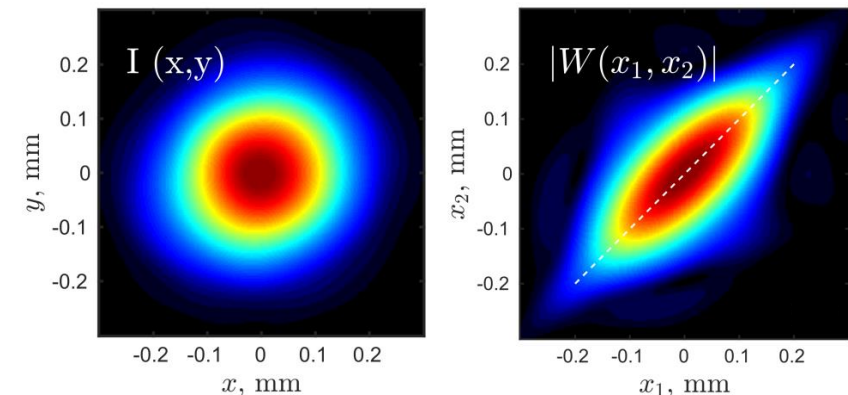
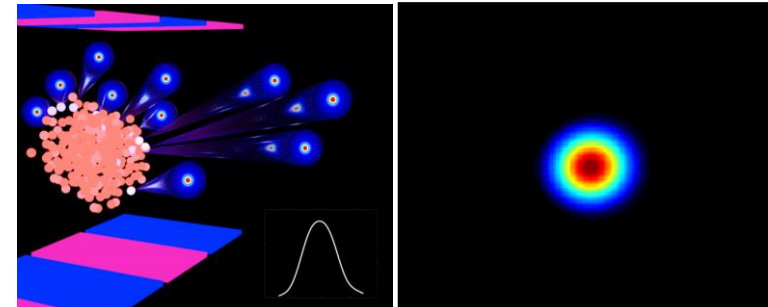
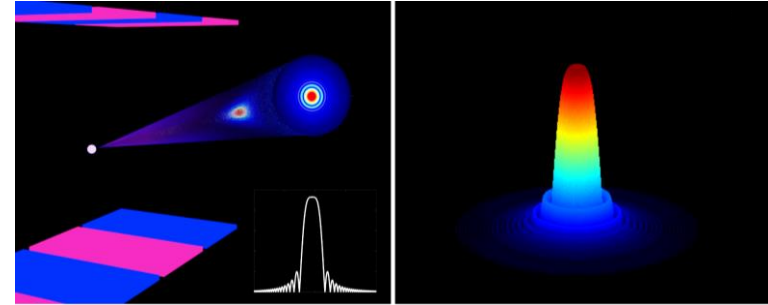
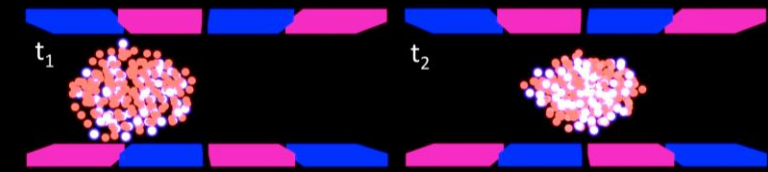
$$\Gamma_t(\mathbf{r}_1, \mathbf{r}_2, t_1, t_2) = \langle E^*(\mathbf{r}_1, t_1) E(\mathbf{r}_2, t_2) \rangle$$



Space-frequency

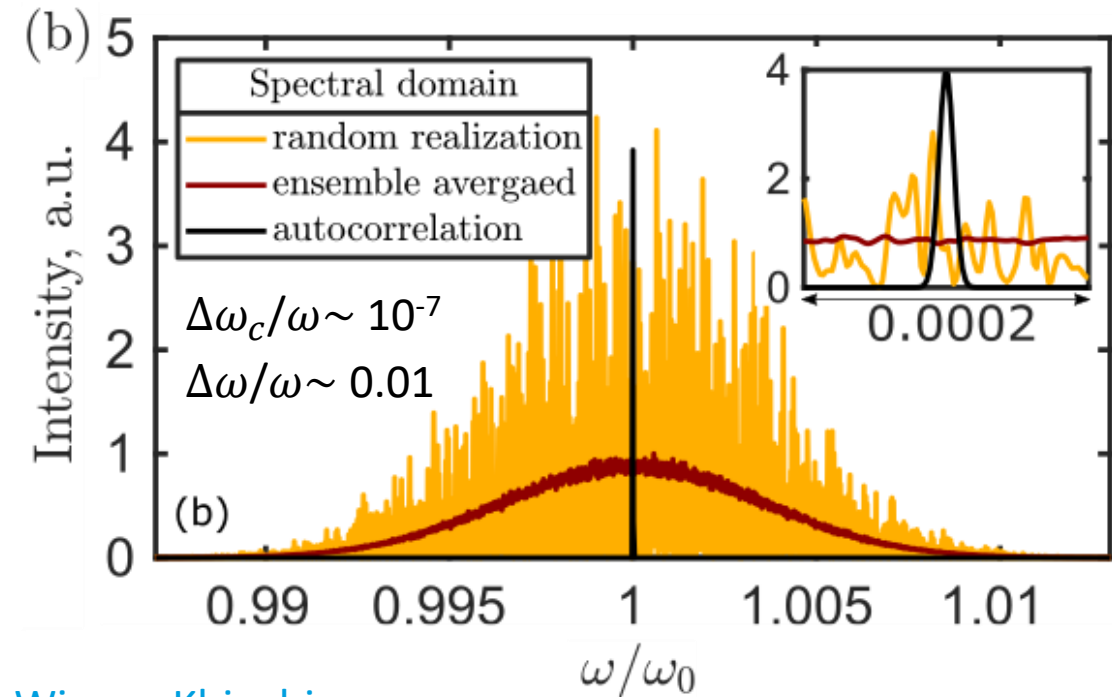
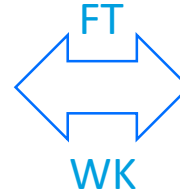
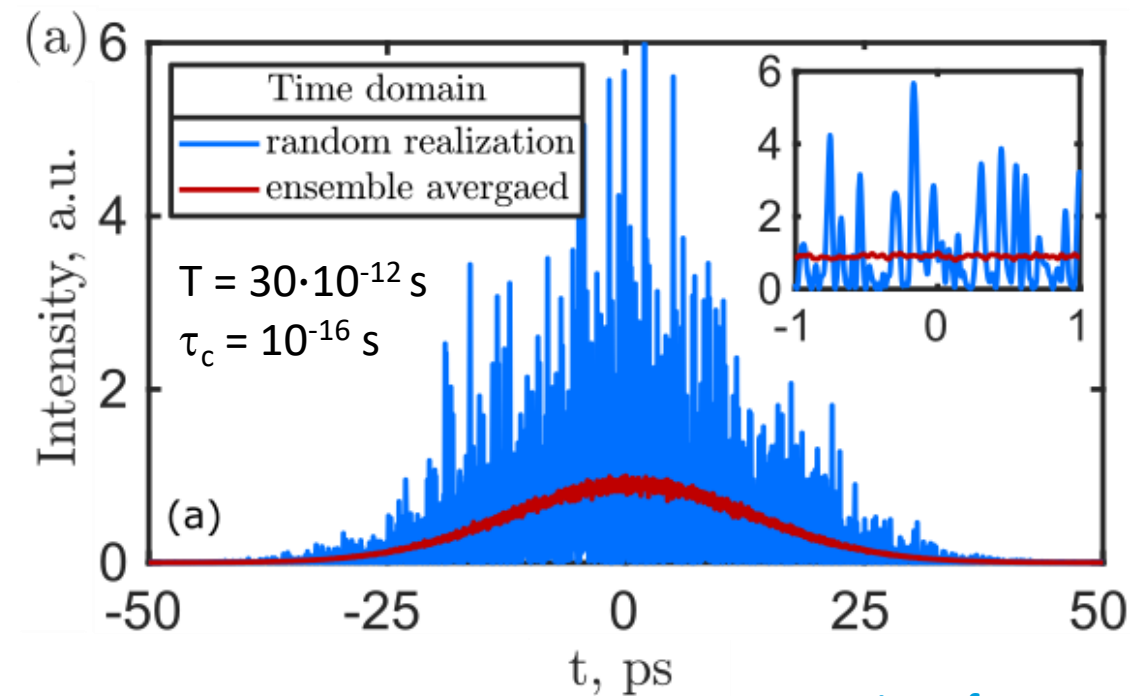
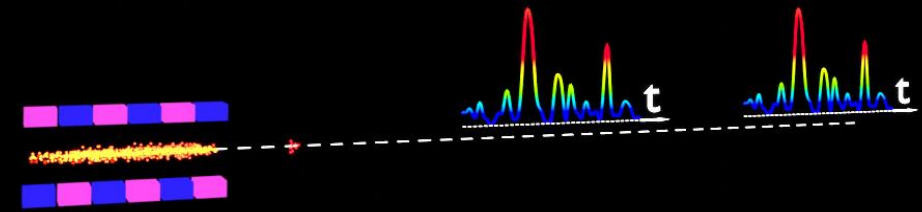
$$\Gamma_\omega(\mathbf{r}_1, \mathbf{r}_2, \omega_1, \omega_2) = \langle E^*(\mathbf{r}_1, \omega_1) E(\mathbf{r}_2, \omega_2) \rangle$$

Spatial and Temporal Coherence are calculated in the frame of statistical optics



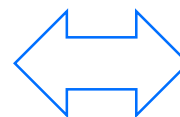
Statistical properties of SR

Pulse structure and Spectrum



Characteristic pulse duration: $T = 30 \cdot 10^{-12} \text{ s}$

Coherence time $\tau_c = 10^{-16} \text{ s}$

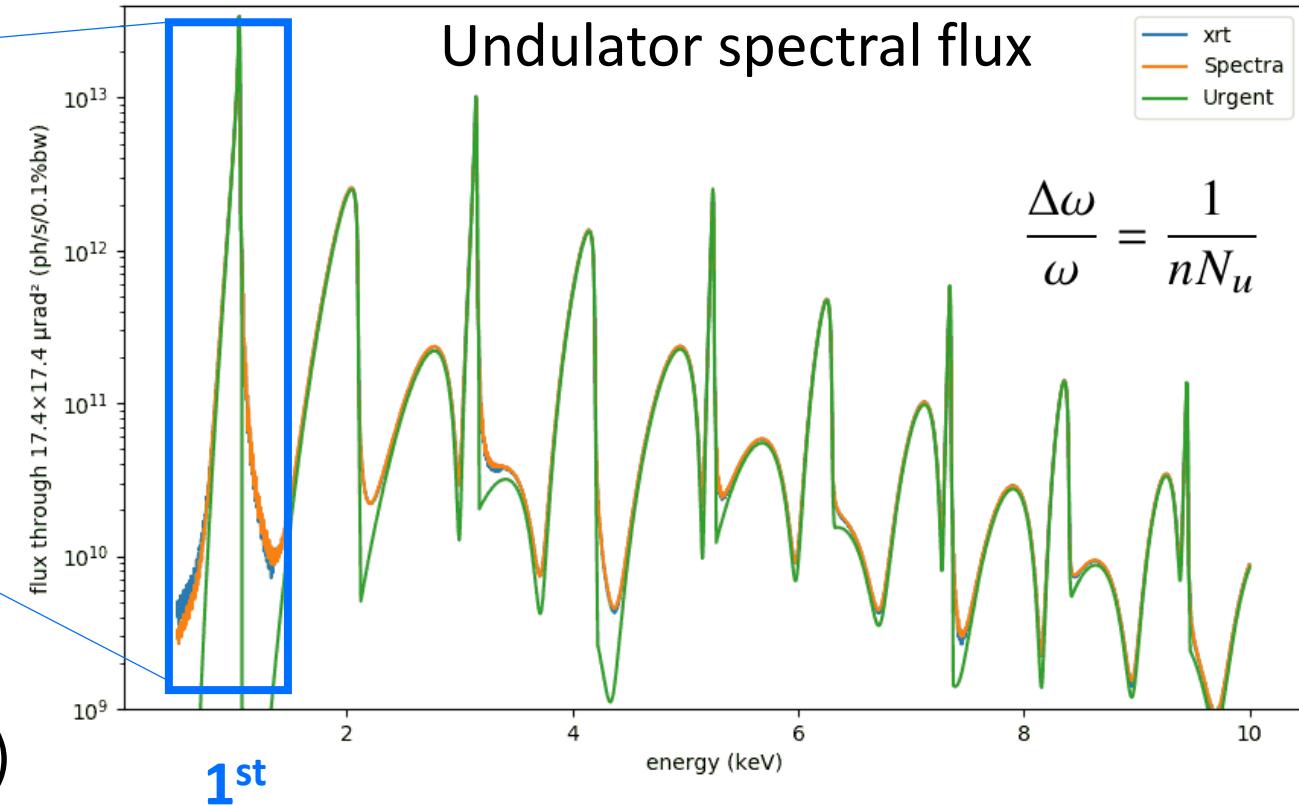
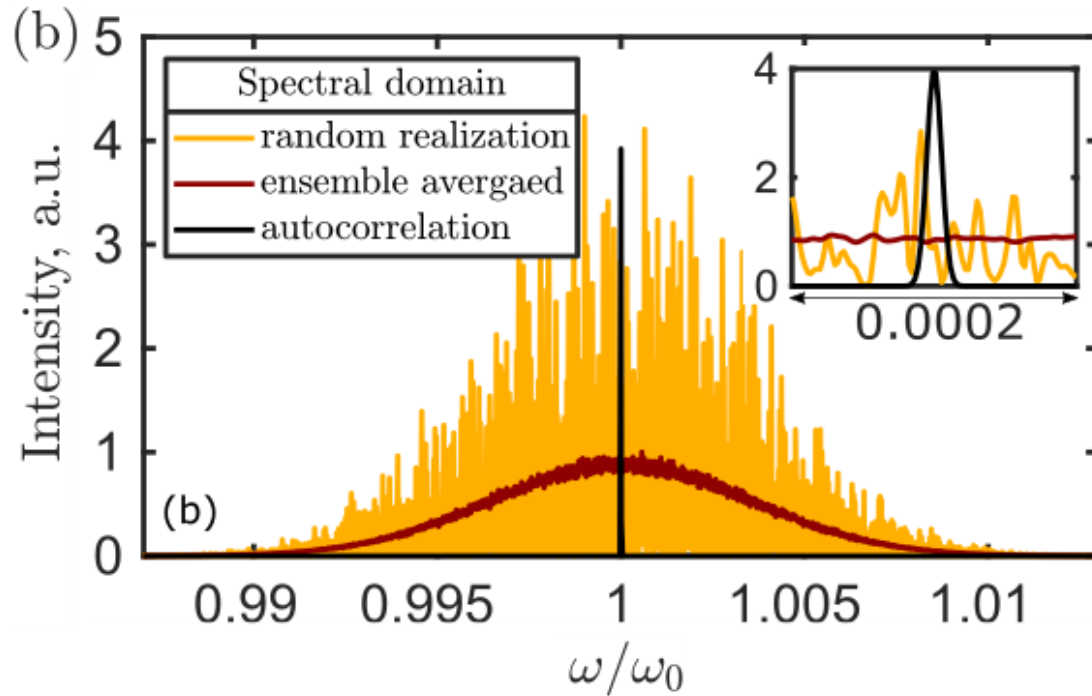
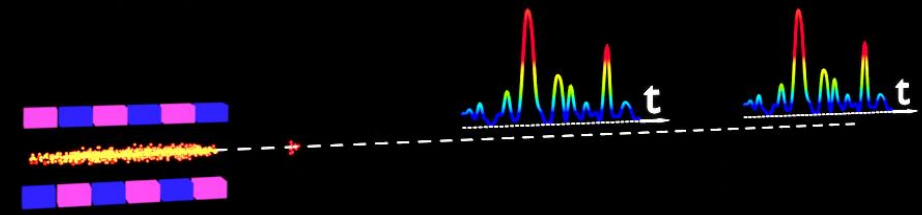


Spectral spike width $\Delta\omega_c/\omega \sim 10^{-7}$

Spectral width of $\Delta\omega/\omega \sim 0.01$

Statistical properties of SR

Spectrum



Spectral width of $\Delta\omega/\omega \sim 0.01$ (10 eV @ 1keV)

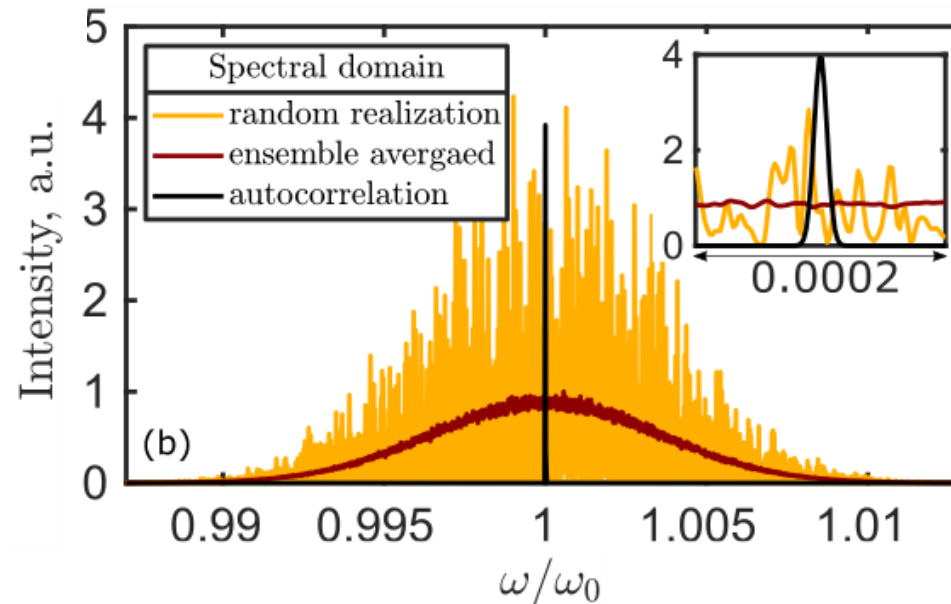
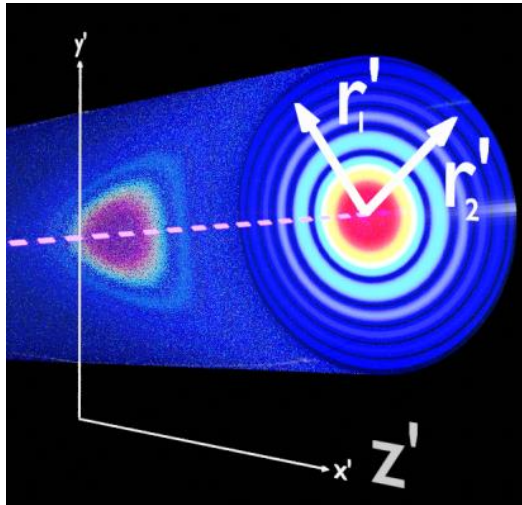
Spectral spike width $\Delta\omega_c/\omega \sim 10^{-7}$

Spatial coherence and correlation functions

Second-order correlation function

Space-frequency

$$\Gamma_{\omega}(\mathbf{r}_1, \mathbf{r}_2, \omega_1, \omega_2) = \langle E^*(\mathbf{r}_1, \omega_1) E(\mathbf{r}_2, \omega_2) \rangle = \underbrace{G_{\perp}(\mathbf{r}_1, \mathbf{r}_2)}_{\text{Cross-Spectral density (CSD)}} \cdot \underbrace{G_{\omega}(\omega_1 - \omega_2)}_{\text{Spectral correlation function}}$$



Cross-Spectral
density (CSD)



Spatial
correlation
Function

Spectral correlation
function



Approximated by
Dirac δ -function
 $\delta(\omega_1 - \omega_2)$

The second-order correlation function in the space-frequency domain is factorized in two correlation functions

Coherence

Cross – Spectral density (CSD) and Spectral Degree of Coherence (SDC)

$$W(\mathbf{r}_1, \mathbf{r}_2, \omega_0) = \langle E(\mathbf{r}_1, \omega) E^*(\mathbf{r}_2, \omega) \rangle \delta(\omega - \omega_0)$$

Intensity:

$$I(\mathbf{r}, \omega) = W(\mathbf{r}, \mathbf{r}, \omega)$$

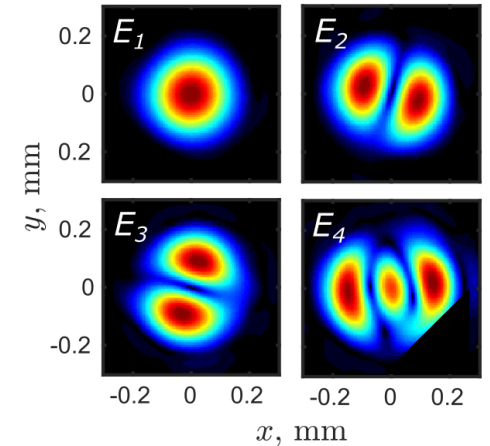
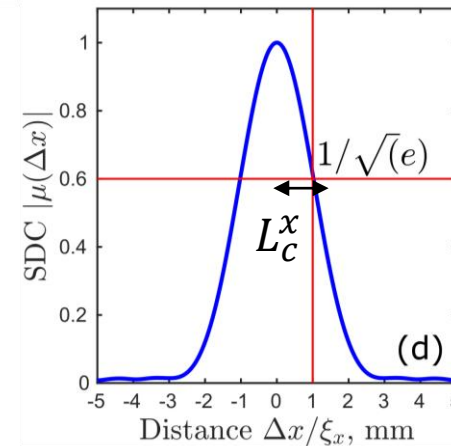
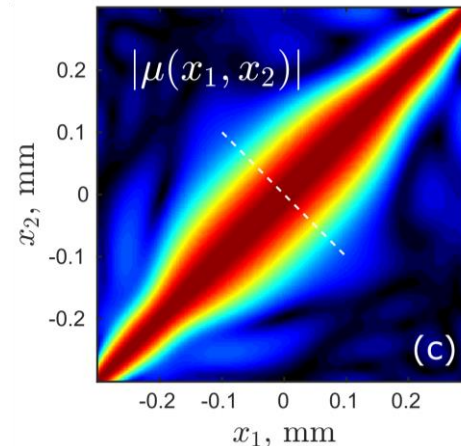
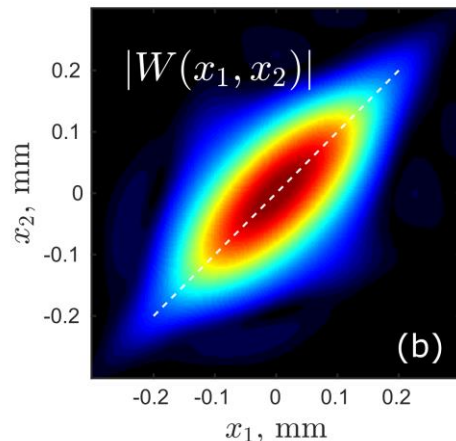
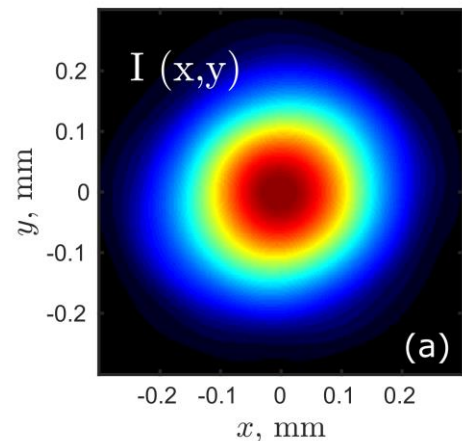
Degree of coherence

$$\zeta = \frac{\iint |W(\mathbf{r}_1, \mathbf{r}_2)|^2 d\mathbf{r}_1 d\mathbf{r}_2}{\int I(\mathbf{r}_1) d\mathbf{r}_1 \int I(\mathbf{r}_2) d\mathbf{r}_2}$$

Incoherent field $0 \leq \zeta \leq 1$ *Coherent field*

Spectral degree of coherence:

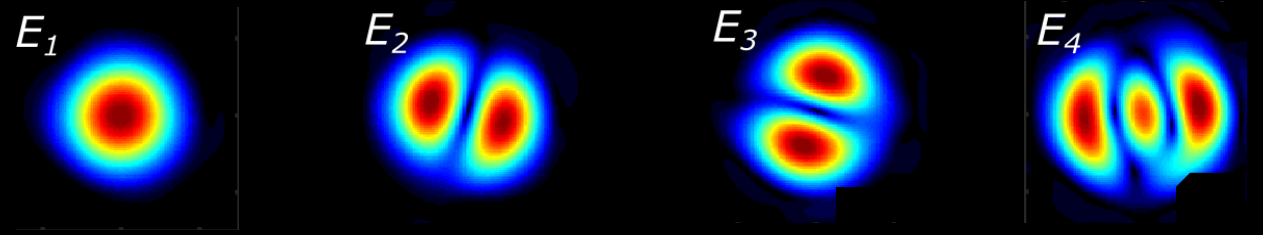
$$\mu(\mathbf{r}_1, \mathbf{r}_2) = \frac{|W(\mathbf{r}_1, \mathbf{r}_2)|}{\sqrt{I(\mathbf{r}_1)} \sqrt{I(\mathbf{r}_2)}}$$



The degree of coherence represents correlation efficiency and characterizes the spatial coherence by a single number

Coherence

Coherent mode representation



Mandel, L. & Wolf, E. (1995). *Optical Coherence and Quantum Optics*.

$$W(\mathbf{r}_1, \mathbf{r}_2, \omega) = \sum_j \beta_j(\omega) E_j^*(\mathbf{r}_1, \omega) E_j(\mathbf{r}_2, \omega)$$

Degree of coherence:

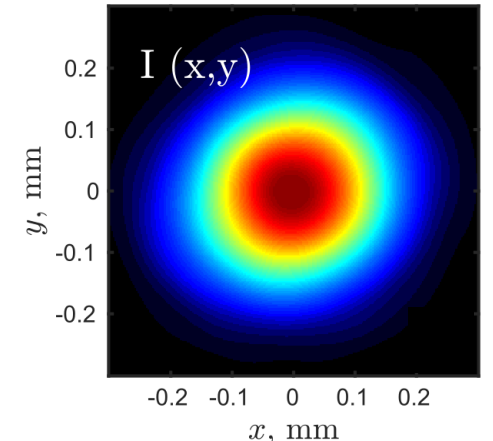
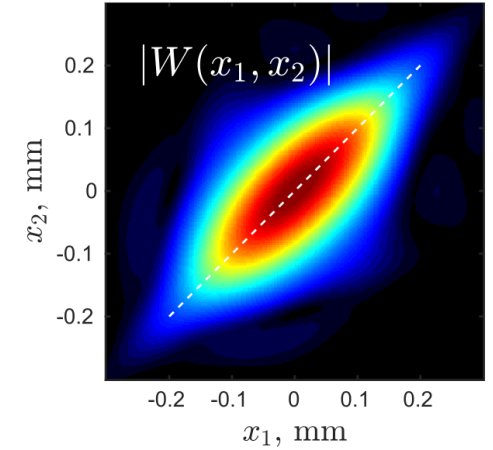
$$\zeta^{DC}(\omega) = \frac{\sum \beta_j^2(\omega)}{(\sum \beta_j(\omega))^2}$$

$$\zeta = \frac{\iint |W(\mathbf{r}_1, \mathbf{r}_2)|^2 d\mathbf{r}_1 d\mathbf{r}_2}{\int I(\mathbf{r}_1) d\mathbf{r}_1 \int I(\mathbf{r}_2) d\mathbf{r}_2}$$

Coherent fraction:

$$\zeta^{CF}(\omega) = \frac{F_{coh}}{F} = \frac{\beta_0(\omega)}{\sum_{j=0}^{\infty} \beta_j(\omega)}$$

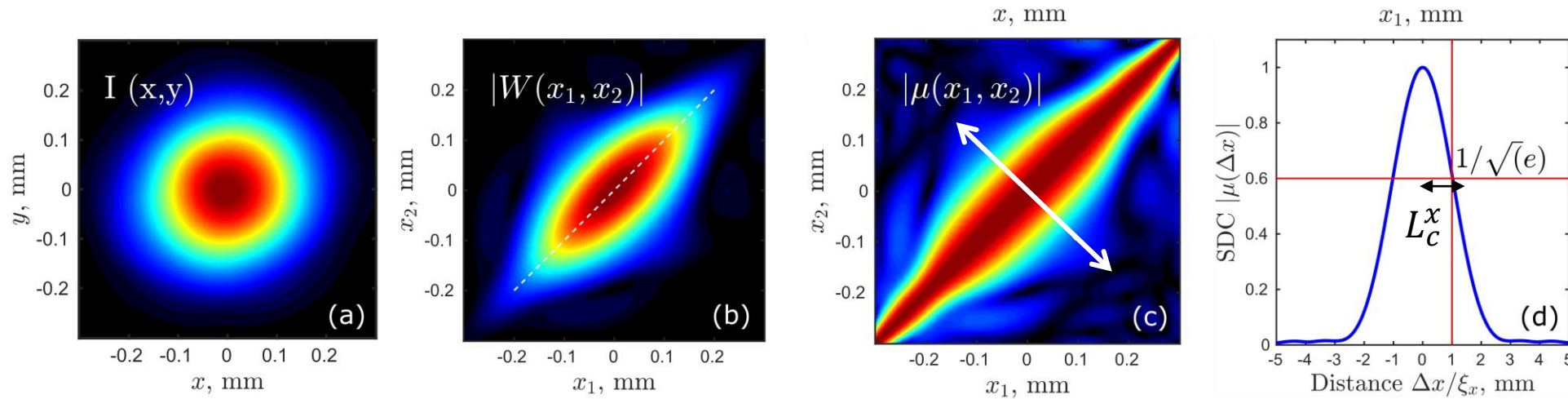
$$B(\mathbf{r}, \mathbf{k}_{\perp}) = W(\mathbf{r}, \mathbf{k}_{\perp}) = \left(\frac{k}{2\pi}\right)^2 \int W_0\left(\mathbf{r} - \frac{1}{2}\Delta\mathbf{r}, \mathbf{r} + \frac{1}{2}\Delta\mathbf{r}\right) e^{i\mathbf{k}_{\perp} \cdot \Delta\mathbf{r}} d\Delta\mathbf{r}$$



Glass, Mark, and Manuel Sanchez del Rio, "Coherent modes of X-ray beams emitted by undulators in new storage rings." *Europhysics Letters* 119.3 (2017)

Manuel Sanchez del Rio, "About the coherent fraction of synchrotron emission." arXiv preprint arXiv:1801.07542 (2018)

Gaussian – Schell Model (GSM)



1. Source is spatially uniform and SDC and Intensity are Gaussian functions:

$$I(\mathbf{r}_1) = I_0 \left[\exp\left(-\frac{x^2}{2\sigma_r^2}\right) \right] \quad W(\mathbf{r}_1, \mathbf{r}_2) = \mu(\mathbf{r}_2 - \mathbf{r}_1) \sqrt{I(\mathbf{r}_1)} \sqrt{I(\mathbf{r}_2)}, \quad \mu(\mathbf{r}_2 - \mathbf{r}_1) = \exp\left(-\frac{(r_{x2} - r_{x1})^2}{2L_x^2} - \frac{(r_{y2} - r_{y1})^2}{2L_y^2}\right)$$

2. Correlation Functions are factorizable into different transverse directions:

$$W(\mathbf{r}_1, \mathbf{r}_2) = W(r_{x1}, r_{x2}) W(r_{y1}, r_{y2}) \quad \zeta^{DC} = \zeta_x^{DC} \zeta_y^{DC}$$

3. Degree of Coherence and Coherent Fraction can be estimated through source parameters:

Degree of coherence:

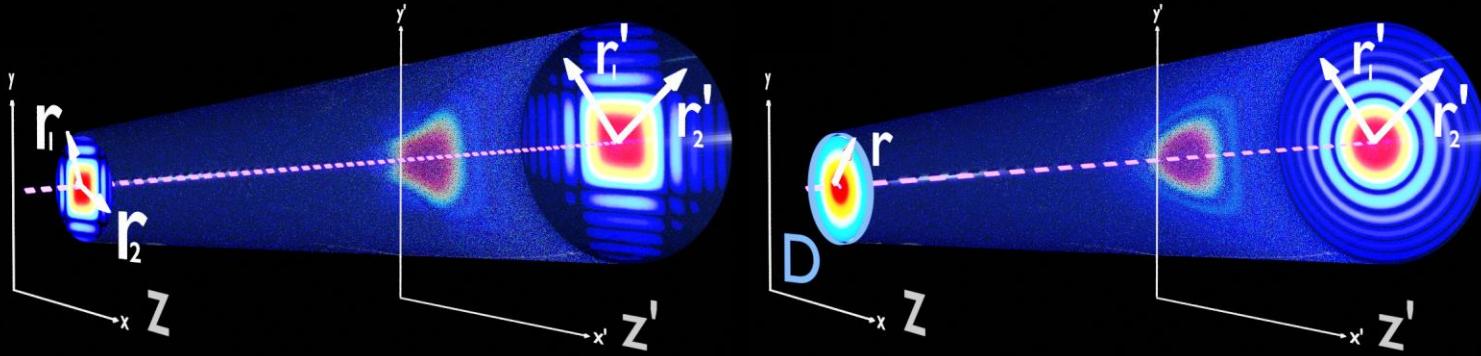
$$\zeta_i^{DC} = 1 / \sqrt{1 + \left(\frac{2\sigma_r}{L_i}\right)^2}$$

Coherent Fraction:

$$\zeta^{CF} \approx \frac{2\zeta_i^{DC}}{1 + \zeta_i^{DC}}$$

Propagation of Correlation functions

Van Cittert-Zernike theorem



Propagation of correlation functions:

$$W(\mathbf{r}_1, \mathbf{r}_2, z_1, \omega) = \int W(\mathbf{r}_1', \mathbf{r}_2', z_0, \omega) P^*(\mathbf{r}_1, \mathbf{r}_1', \omega) P(\mathbf{r}_2, \mathbf{r}_2', \omega) d\mathbf{r}_1' d\mathbf{r}_2'$$

Propagator:

$$P(\mathbf{r}, \mathbf{r}', \omega) = \frac{-ik}{2\pi} \frac{e^{ikz}}{z} \exp\left(ik \frac{|\mathbf{r} - \mathbf{r}'|^2}{2z}\right)$$

Incoherent source MOI function: $J(\mathbf{r}_1, \mathbf{r}_2) = I(\mathbf{r}_1) \delta(\mathbf{r}_2 - \mathbf{r}_1)$



Coherent Length :

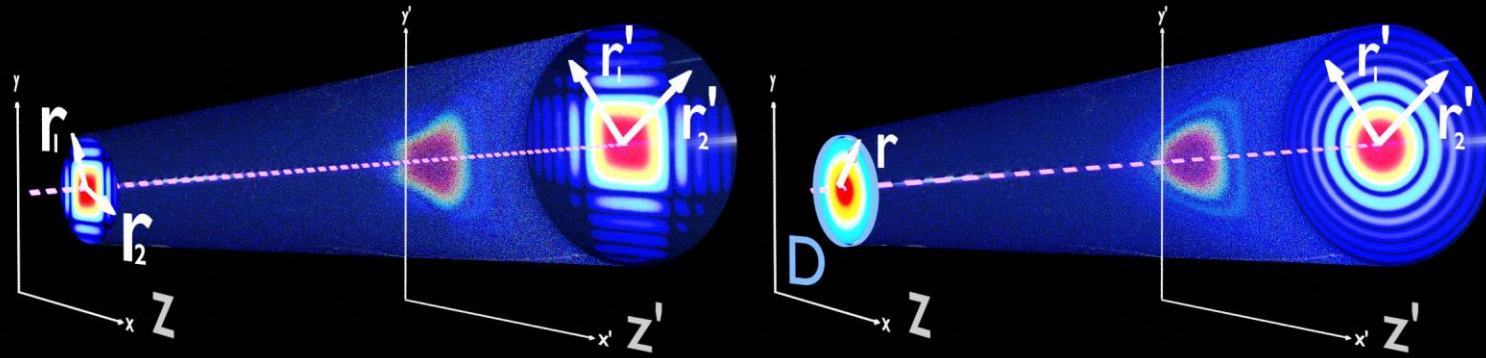
$$L_c \sim \frac{\lambda z}{D}$$

Normalized CSD:

$$\mu(\mathbf{r}_1', \mathbf{r}_2') = \frac{\exp(-i\Psi) \int I(\mathbf{r}) \exp\left(-\frac{i2\pi}{\lambda z} (\mathbf{r}_2' - \mathbf{r}_1') \cdot \mathbf{r}\right) d\mathbf{r}}{\int I(\mathbf{r}) d\mathbf{r}}$$

Propagation of Correlation functions

Van Cittert-Zernike theorem



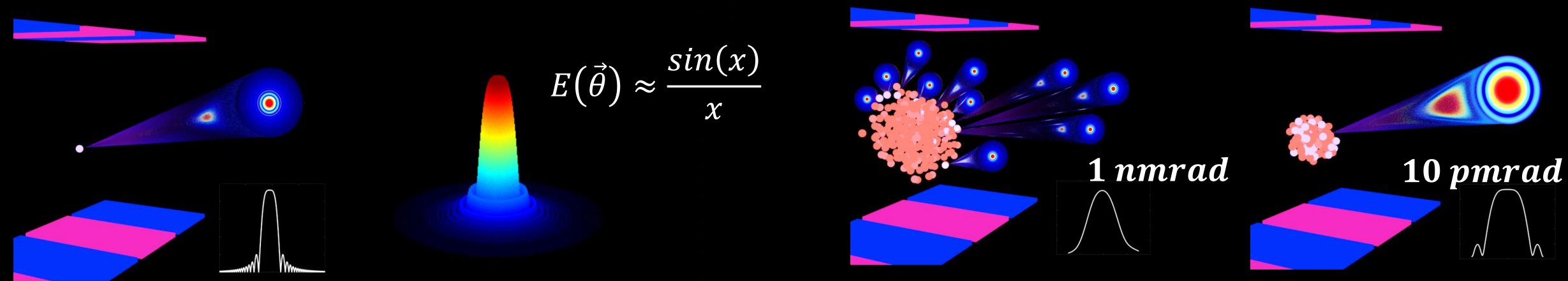
Coherent Length :

$$L_c \sim \frac{\lambda z}{D}$$



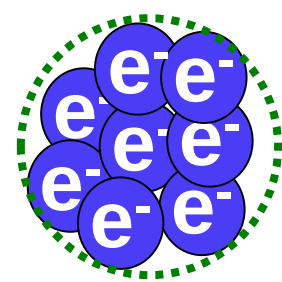
Statistical properties of SR – Diffraction Limit

Radiation Profile



Electron Emittance:

$$\epsilon_{e^-} = \sigma_e \cdot \sigma_{e'}$$

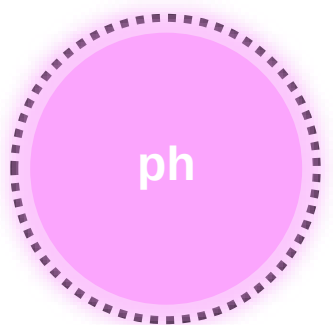


4th Generation SR source

10 pm rad

Photon Emittance (single emission, coherent):

$$\epsilon_r = \sigma_r \cdot \sigma_{r'}$$



$$\equiv \frac{\lambda}{4\pi}$$

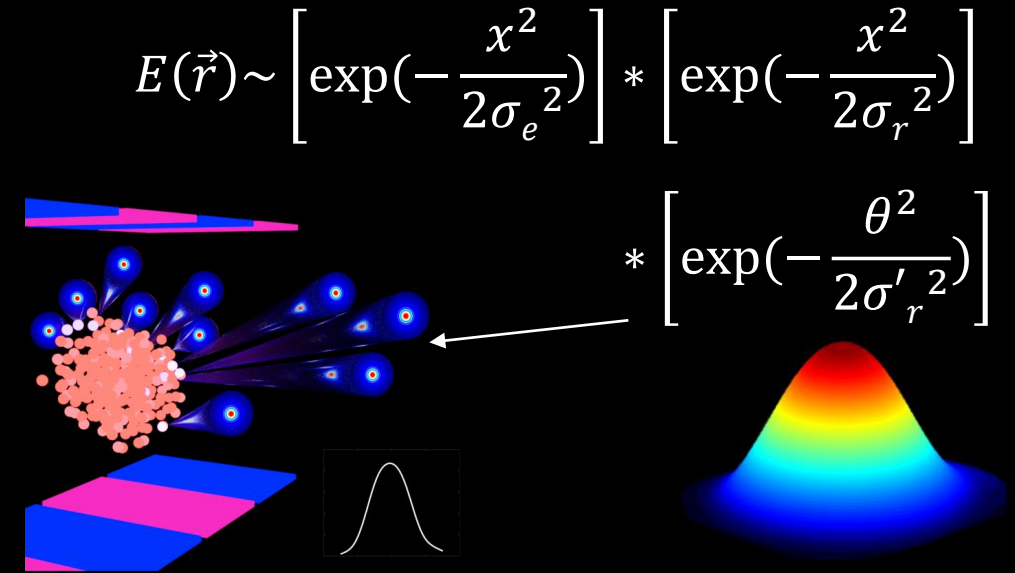
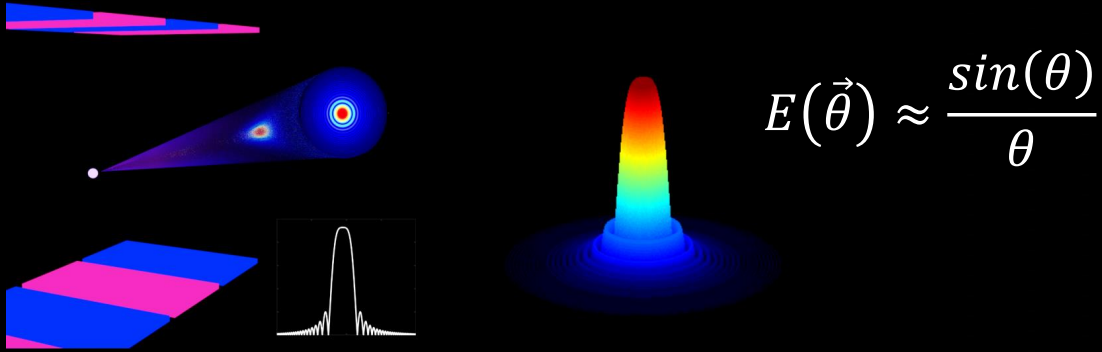
Photon Emittance

197 pm rad	500 eV
8.2 pm rad	12 keV
1.9 pm rad	50 keV

$$\epsilon_{coh} = \Delta x \cdot \Delta \theta \geq \lambda/4\pi$$

Statistical properties of SR

Gaussian Approximation



Coherent phase-space area: $\varepsilon_r = \sigma_r \cdot \sigma_r' = \lambda/4\pi$

Total Photon Emittance: $\varepsilon_{ph} = \Sigma_{ph} \cdot \Sigma'_{ph}$

Source size: $\Sigma_{ph} \approx [\sigma_e^2 + \sigma_r^2]^{1/2}$

Photon Divergence: $\Sigma'_{ph} \approx [\sigma_e'^2 + \sigma_r'^2]^{1/2}$

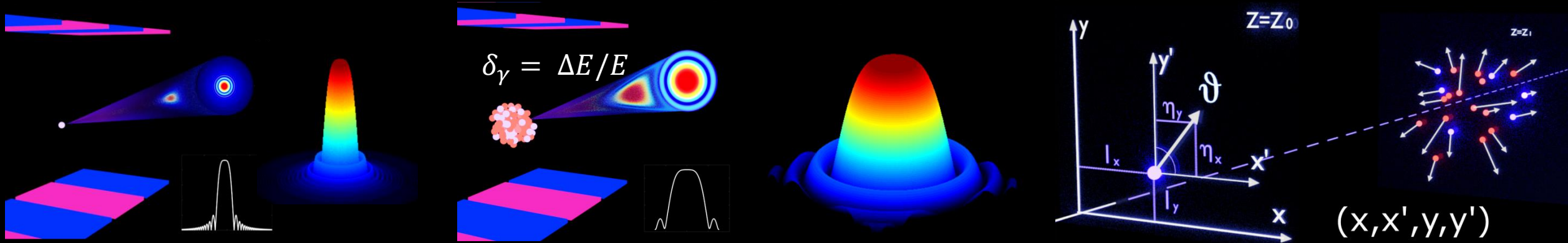
Coherent Fraction:

$$\zeta^{CF}(\omega) = \frac{F_{coh}}{F} = \frac{\varepsilon_{coh}}{\varepsilon_{ph}} = \frac{\lambda/4\pi}{\Sigma_{ph} \cdot \Sigma'_{ph}}$$

Single electron parameters:

$$\sigma_r = \frac{\sqrt{2\lambda_n L_u}}{4\pi} \quad \sigma_r' = \sqrt{\frac{\lambda_n}{2L_u}}$$

Diffraction Limit and Energy Spread



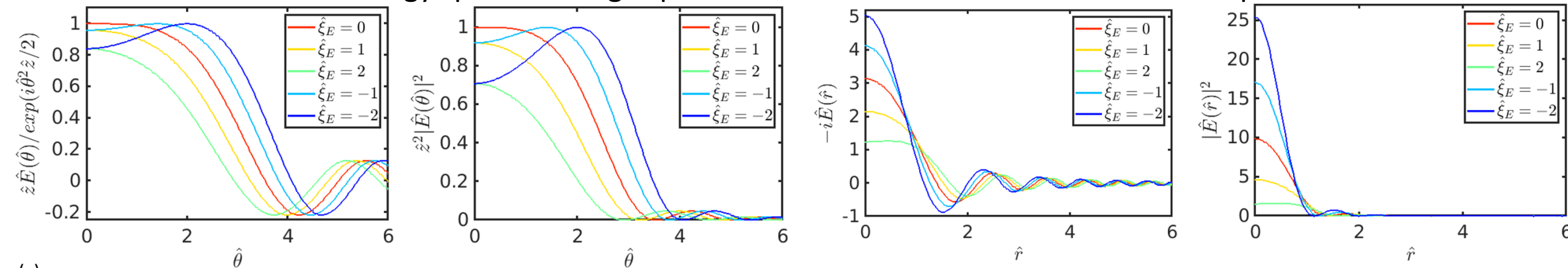
Far field amplitude distribution of a single electron:

$$E(\boldsymbol{\theta}) \sim \exp \left[i \frac{\omega}{c} \left(\frac{z\theta^2}{2} - \boldsymbol{\theta} \cdot \mathbf{l} \right) \right] \text{sinc} \left[\frac{2\pi N_u (\gamma - \gamma_1)}{\gamma_1} + \frac{\omega L_u |\boldsymbol{\theta} - \boldsymbol{\eta}|^2}{4c} \right]$$

Far field amplitude of the electron bunch:

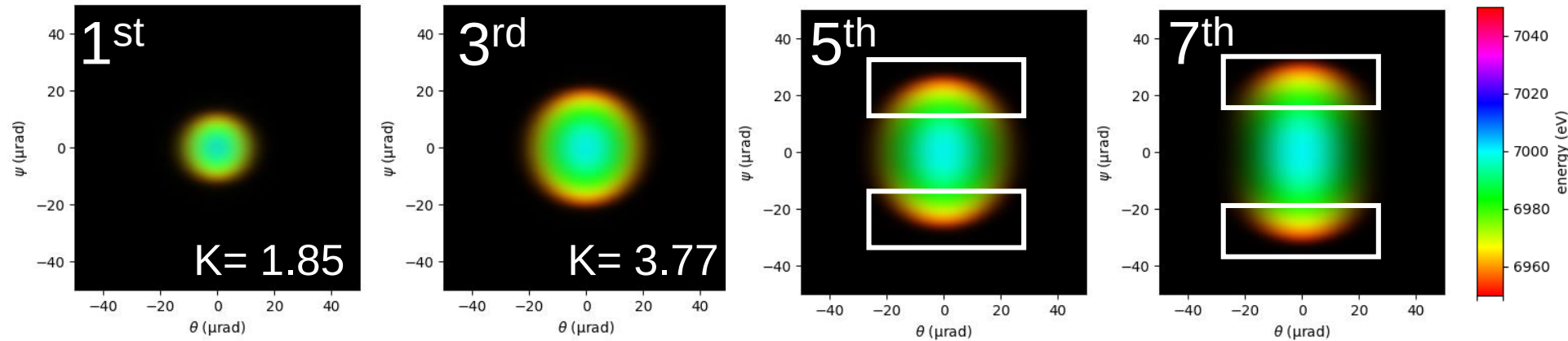
$$E_{total}(\boldsymbol{\theta}) = \int E(\boldsymbol{\theta}) f(\mathbf{l}) f(\boldsymbol{\eta}) f(\gamma_E) d\boldsymbol{\theta} d\mathbf{l} d\boldsymbol{\eta} d\gamma_E$$

Energy spread enlarges photon beam and transforms the beam shape

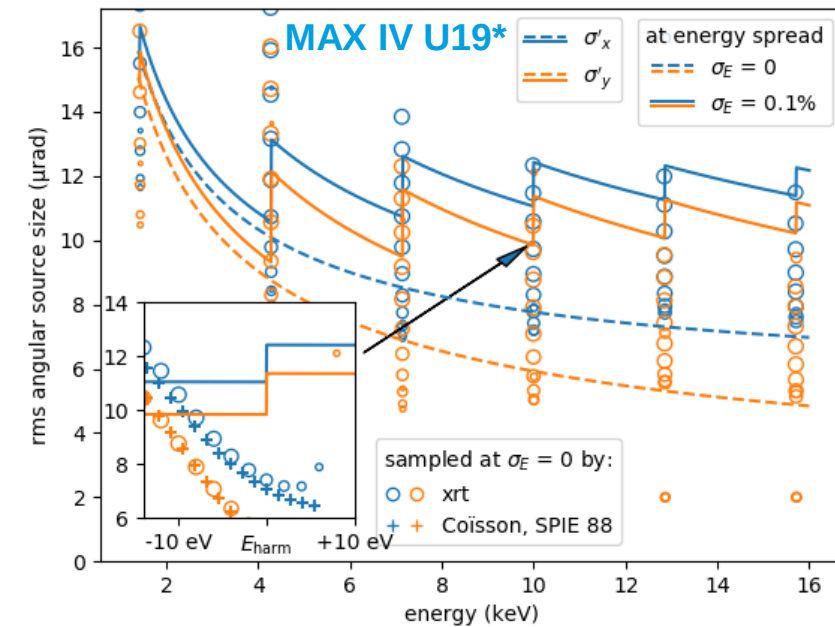
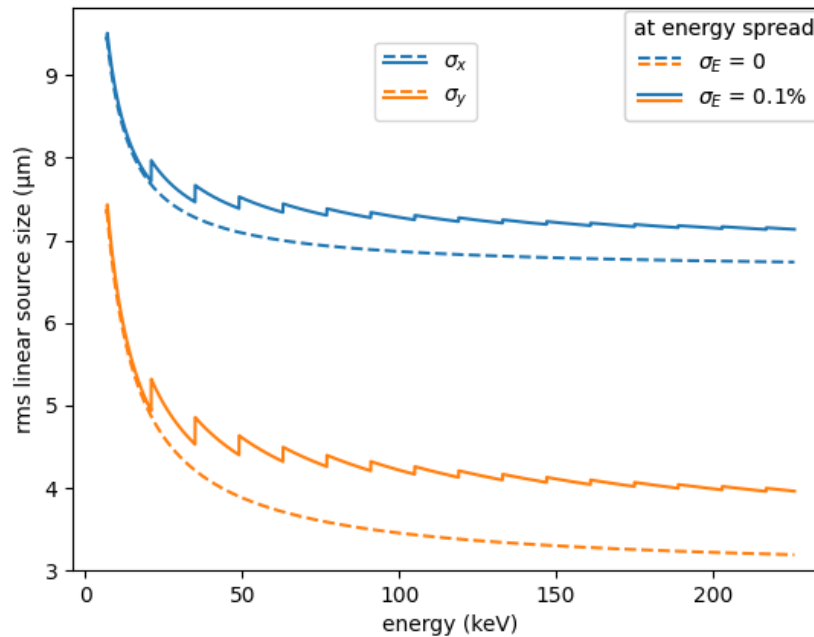
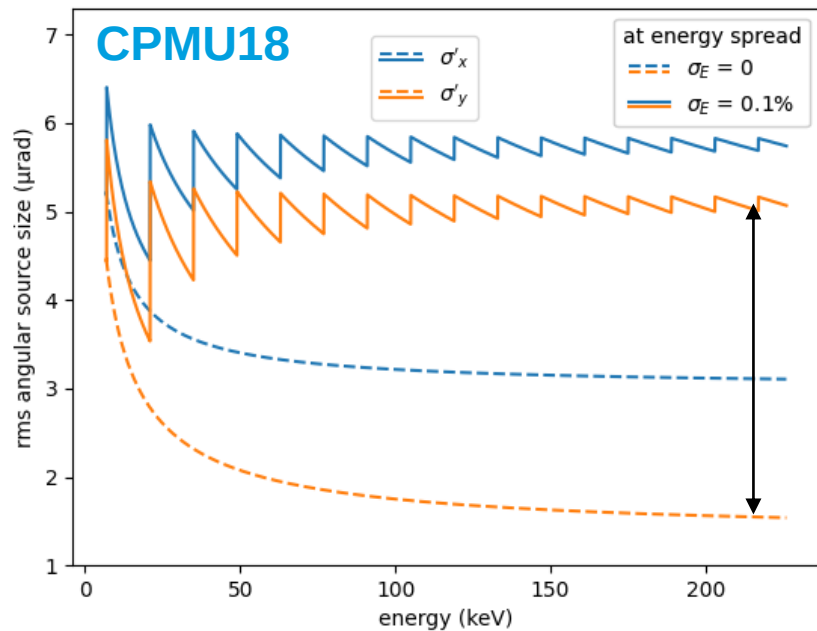


Energy Spread

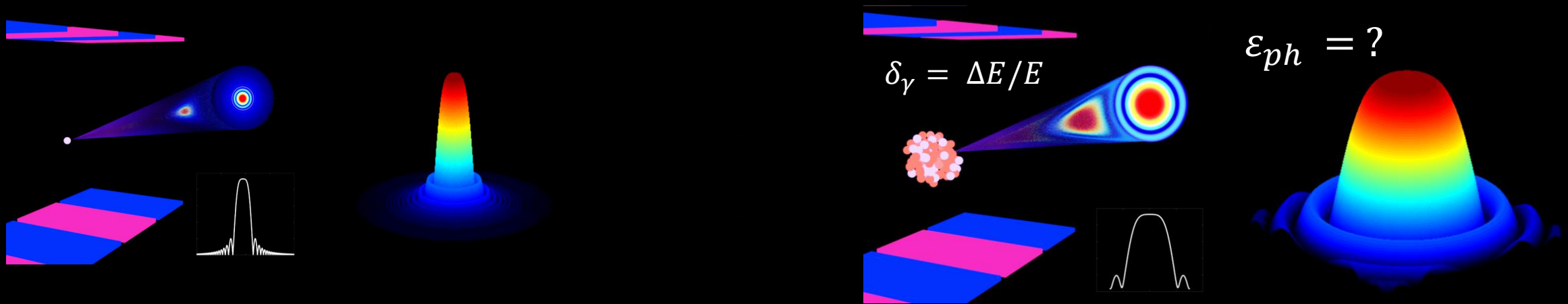
Electron Energy 6.0 GeV
Beta-function, β_x, β_y 2.2 m
Electron beam emittance, ε_e 20/4 pmrad
Energy spread 0, 0.1%



For very low emittance beam, the e-beam energy spread starts to play a role at higher harmonics



Diffraction Limit and Energy Spread



Source size:

$$\Sigma_{ph} \approx [\sigma_e^2 + \sigma_r^2 * 4 * N(\delta_\gamma/4\delta_n)^{2/3}]^{1/2}$$

Source divergence:

$$\Sigma'_{ph} \approx [\sigma'_e{}^2 + \sigma'_r{}^2 * N(\delta_\gamma/\delta_n)]^{1/2}$$

In the diffraction limit:

$$\Sigma_{ph} \approx [\cancel{\sigma_e^2} + \sigma_r^2 * 4]^{1/2}$$

$$\Sigma'_{ph} \approx [\cancel{\sigma'_e{}^2} + \sigma'_r{}^2]^{1/2}$$



$$\sigma_r = \frac{\sqrt{2\lambda_n L_u}}{4\pi} \quad \sigma'_r = \sqrt{\frac{\lambda_n}{2L_u}}$$



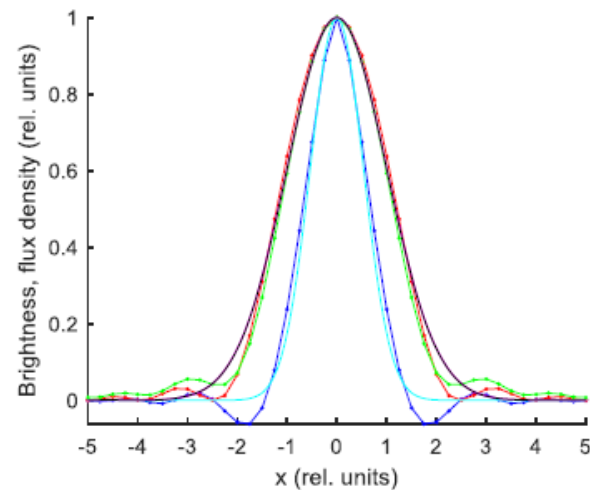
$$\epsilon_{coh} = \frac{\lambda}{2\pi}$$

Methods overview

Method 1

1D section through the brightness distribution

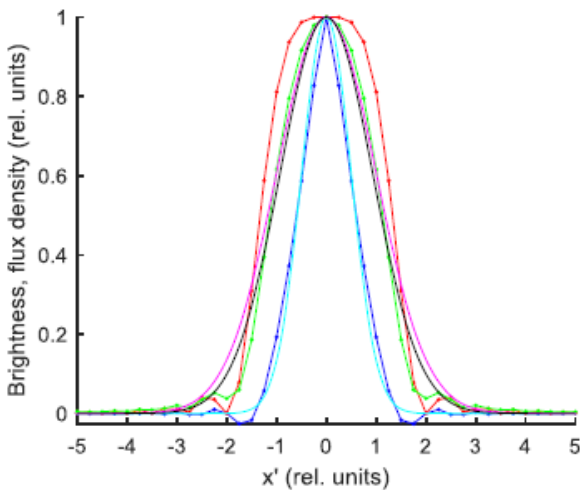
$$\Sigma_{ph\ x} = \frac{\int B(x, 0, 0, 0) dx}{\int \sqrt{2\pi} B(x, 0, 0, 0)_{max}}$$



Method 2

3D projection + 1D Gaussian fit

$$dF/dx = \iiint B(x, x', y, y') dx' dy' dy$$
$$dF/dx' = \iiint B(x, x', y, y') dx dy dy'$$



Method 3

2D projection

$$\Sigma_{ph\ x,y} = \left(\frac{F}{2\pi(dF/dS)_{max}} \right)$$

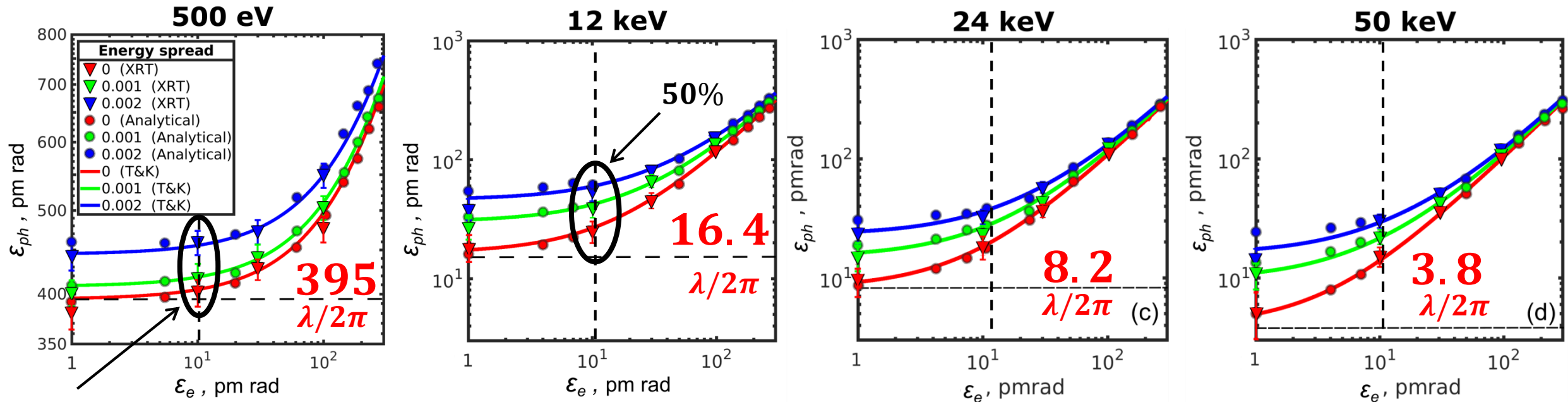
$$\Sigma'_{ph\ x,y} = \left(\frac{F}{2\pi(dF/d\Omega)_{max}} \right)$$

$$\frac{dF}{dS} = \iint B(x, x', y, y') dx' dy' \quad \frac{dF}{d\Omega} = \iint B(x, x', y, y') dx dy$$

Model	$\sigma_{R'}$	σ_R	ε_R	β_R
1	$\sqrt{\lambda/L}$	$\sqrt{\lambda L}/4\pi$	$\lambda/4\pi$	$L/4\pi$
2	$\sqrt{\lambda/2L}$	$\sqrt{2\lambda L}/4\pi$	$\lambda/4\pi$	$L/2\pi$
3	$\sqrt{\lambda/2L}$	$\sqrt{2\lambda L}/2\pi$	$\lambda/2\pi$	L/π
4	$\sqrt{\lambda/4L}$	$\sqrt{\lambda L}/2\pi$	$\lambda/4\pi$	L/π

FIG. 3. Spatial (left) and angular (right) distributions of the brightness function; 1D sections (blue line), sections of the 2D projected flux density (red line), and full 3D projections (green line) along the respective axes, with respective Gaussian fits (cyan, magenta, and black lines); magenta and black curves are very close together.

Photon Emittance



12% difference of the photon emittance due to the energy spread effect

Source size:

$$\Sigma_{ph} \approx [\sigma_e^2 + \sigma_r^2 * 4 * N(\delta_\gamma/4\delta_n)^{2/3}]^{1/2}$$

Total Photon Emittance:

$$\epsilon_{ph} = \Sigma_{ph} \cdot \Sigma'_{ph}$$

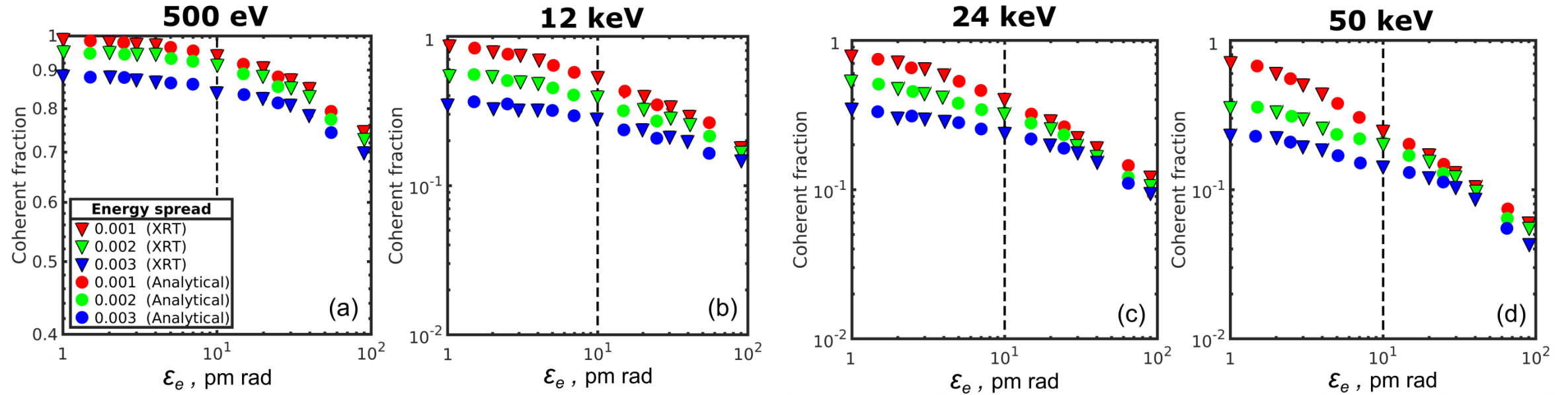
Source divergence:

$$\Sigma'_{ph} \approx [\sigma_e'^2 + \sigma_r'^2 * N(\delta_\gamma/\delta_n)]^{1/2}$$

Coherent phase-space area:

$$\epsilon_r = \sigma_r \cdot \sigma_r' = \lambda/2\pi$$

Coherent Fraction



Independent mode decomposition:

$$\zeta^{CF}(\omega) = \frac{F_{coh}}{F} = \frac{\beta_0(\omega)}{\sum_{j=0}^{\infty} \beta_j(\omega)}$$

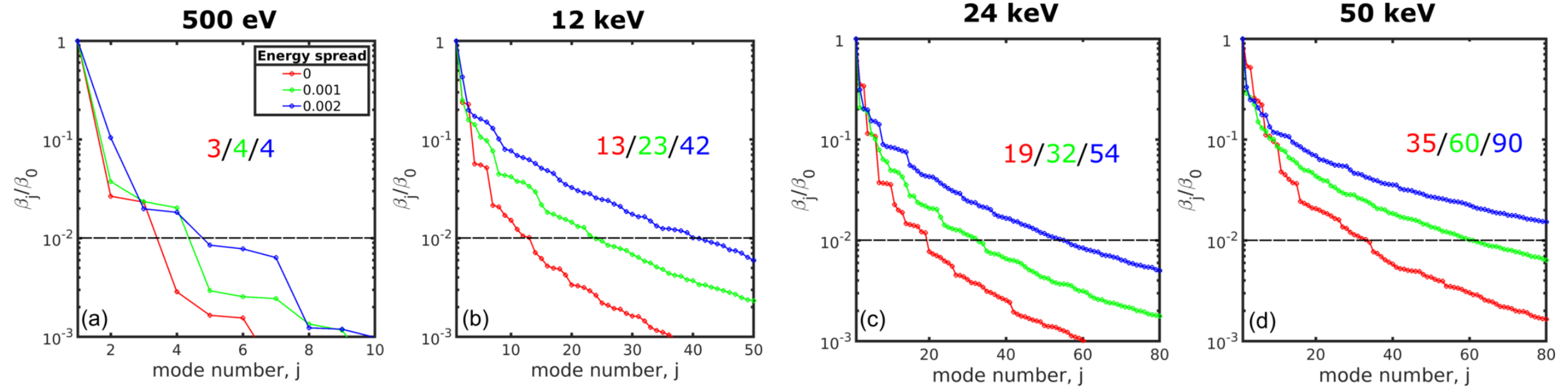
Simple analytics:

$$\zeta^{CF}(\omega) = \frac{\epsilon_{coh}}{\epsilon_{ph}} = \frac{\lambda/2\pi}{\Sigma_{ph} \cdot \Sigma'_{ph}}$$

$$\Sigma_{ph} \approx [\sigma_e^2 + \sigma_r^2 * 4 * N(\delta_\gamma/4\delta_n)^{2/3}]^{1/2}$$

$$\Sigma'_{ph} \approx [\sigma'_e{}^2 + \sigma'_r{}^2 * N(\delta_\gamma/\delta_n)]^{1/2}$$

Coherent Fraction



Independent mode decomposition:

$$\zeta^{CF}(\omega) = \frac{F_{coh}}{F} = \frac{\beta_0(\omega)}{\sum_{j=0}^{\infty} \beta_j(\omega)}$$

Simple analytics:

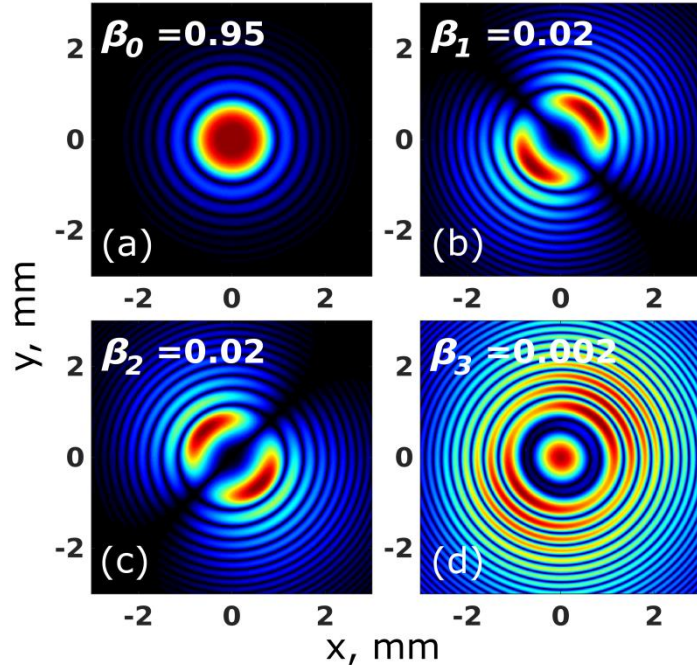
$$\zeta^{CF}(\omega) = \frac{\varepsilon_{coh}}{\varepsilon_{ph}} = \frac{\lambda/2\pi}{\Sigma_{ph} \cdot \Sigma'_{ph}}$$

$$\Sigma_{ph} \approx [\sigma_e^2 + \sigma_r^2 * 4 * N(\delta_\gamma/4\delta_n)^{2/3}]^{1/2}$$

$$\Sigma'_{ph} \approx [\sigma'_e{}^2 + \sigma'_r{}^2 * N(\delta_\gamma/\delta_n)]^{1/2}$$

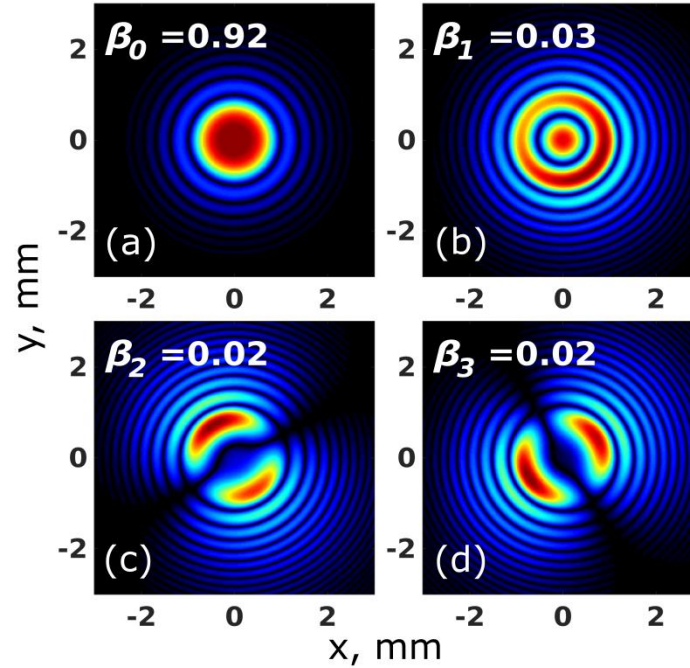
Coherent mode representation – 500 eV

zero relative energy spread



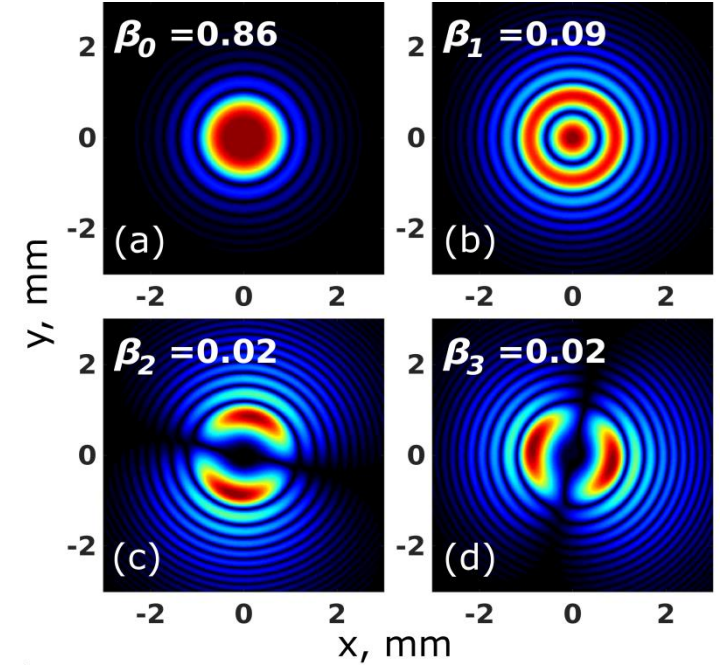
$$\zeta^{DC} = 90\%$$

10^{-3} energy spread



$$\zeta^{DC} = 84\%$$

$2 \cdot 10^{-3}$ energy spread



$$\zeta^{DC} = 74\%$$

Degree of coherence:

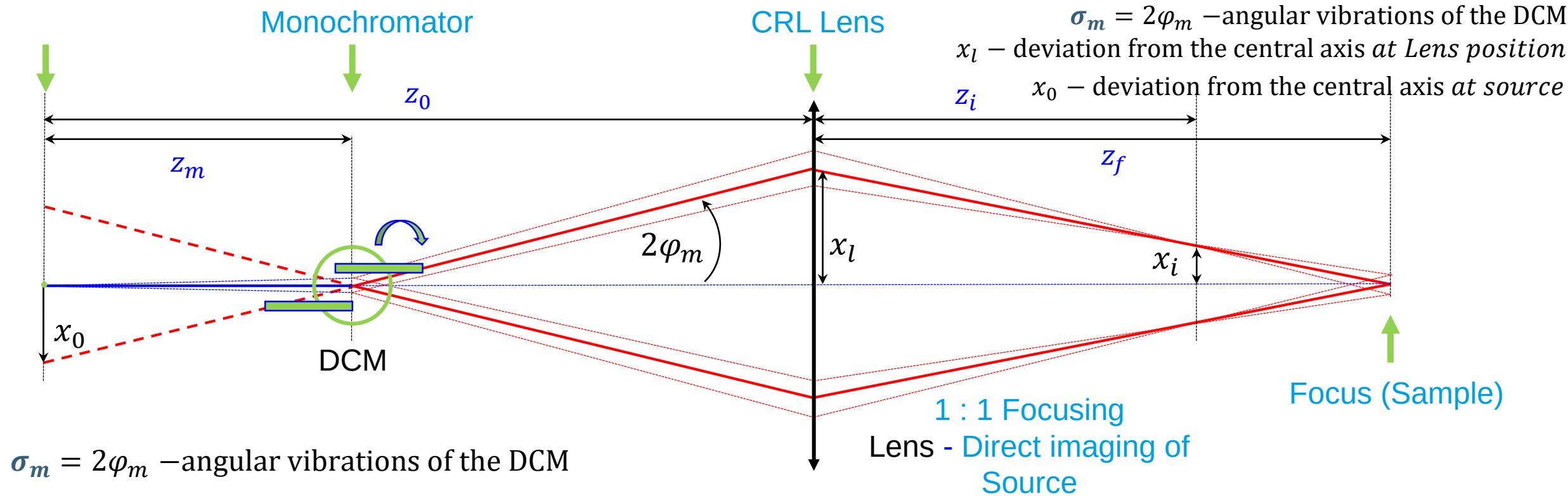
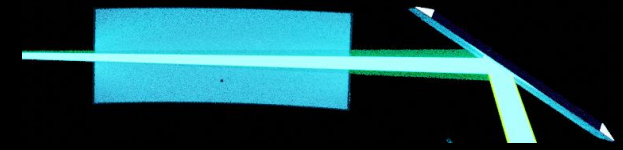
$$\zeta^{DC}(\omega) = \frac{\sum \beta_j^2(\omega)}{(\sum \beta_j(\omega))^2}$$

Coherent fraction:

$$\zeta^{CF}(\omega) = \frac{F_{coh}}{F} = \frac{\beta_0(\omega)}{\sum_{j=0}^{\infty} \beta_j(\omega)}$$

True diffraction limit at high-energy 4th generation source (10 pm rad) will be reached at soft X-ray energies

Vibration – Source and Optics



$\sigma_m = 2\varphi_m$ – angular vibrations of the DCM

$$\Sigma_0 = \sqrt{\Sigma_x^2 + \Sigma_S^2 + \sigma_m^2 z_m^2}$$

$$\Sigma'_0 = \sqrt{\Sigma'_x{}^2 + \Sigma'_S{}^2 + \sigma_m^2}$$

$$\Sigma_L = \sqrt{\Sigma_x^2 + \Sigma_S^2 + \Sigma'_x{}^2 z_0^2 + \Sigma'_S{}^2 z_0^2 + \sigma_m^2 (z_0 - z_m)^2}$$

Σ_x – source size

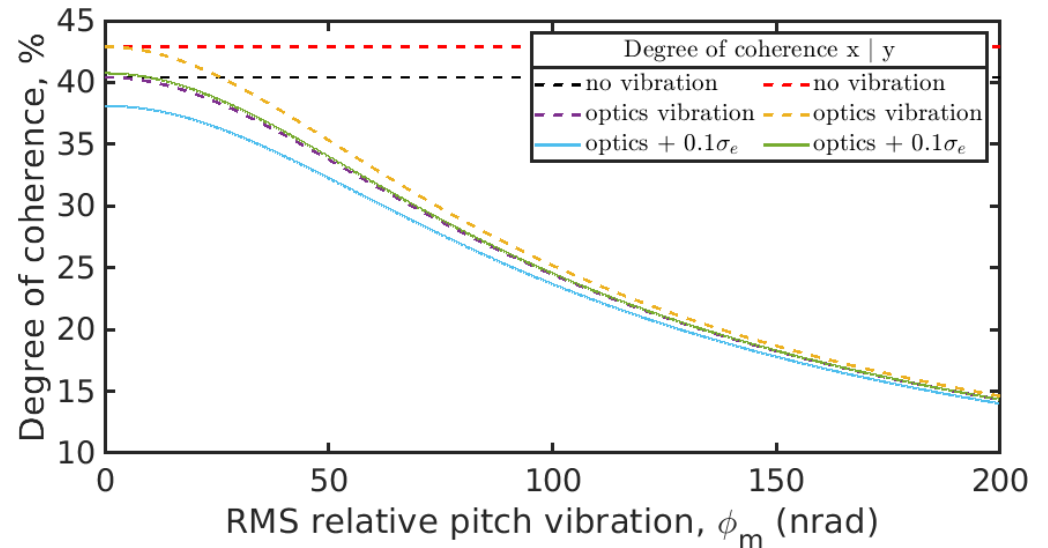
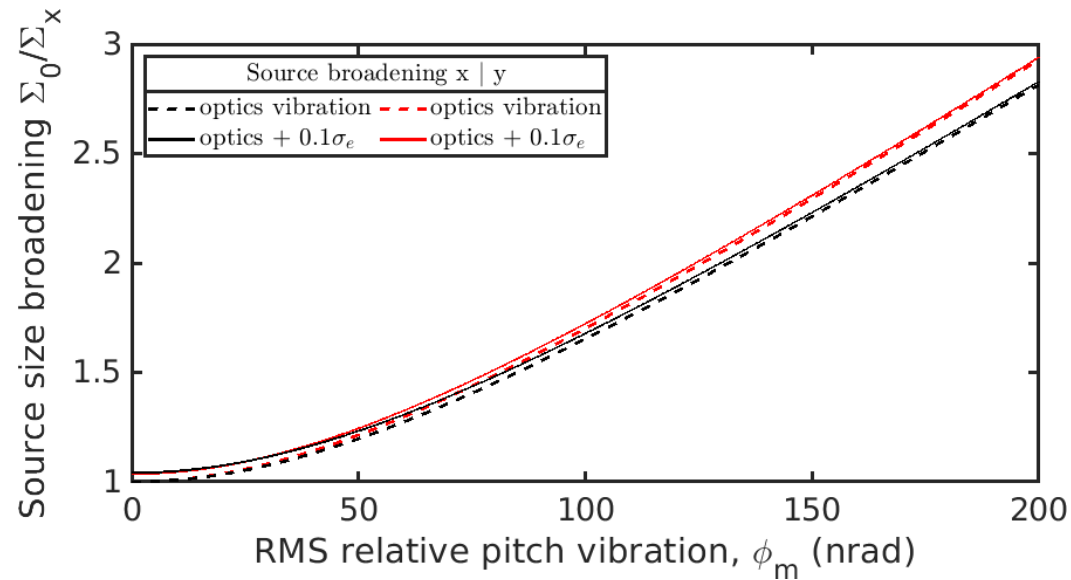
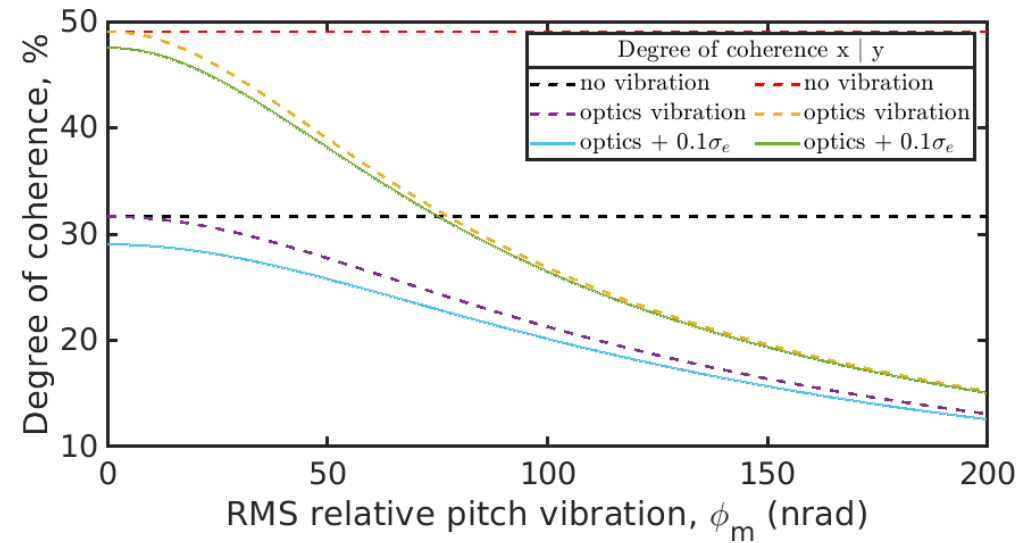
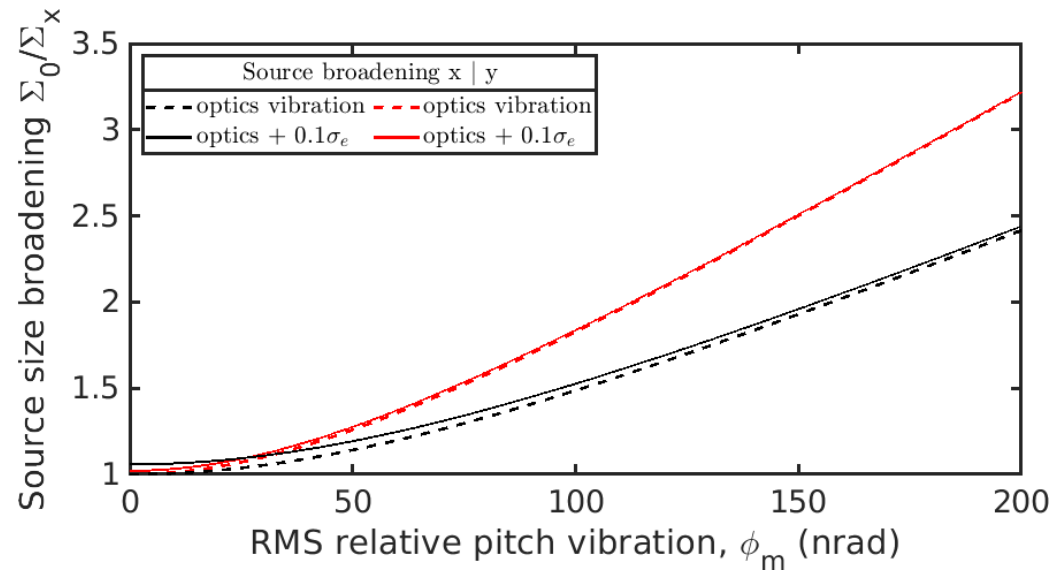
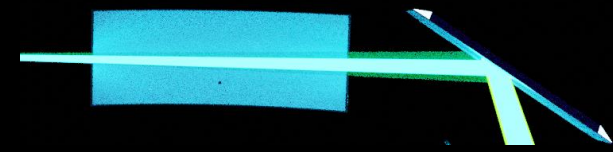
Σ_S – translational movement of the source

Σ'_x – source divergence

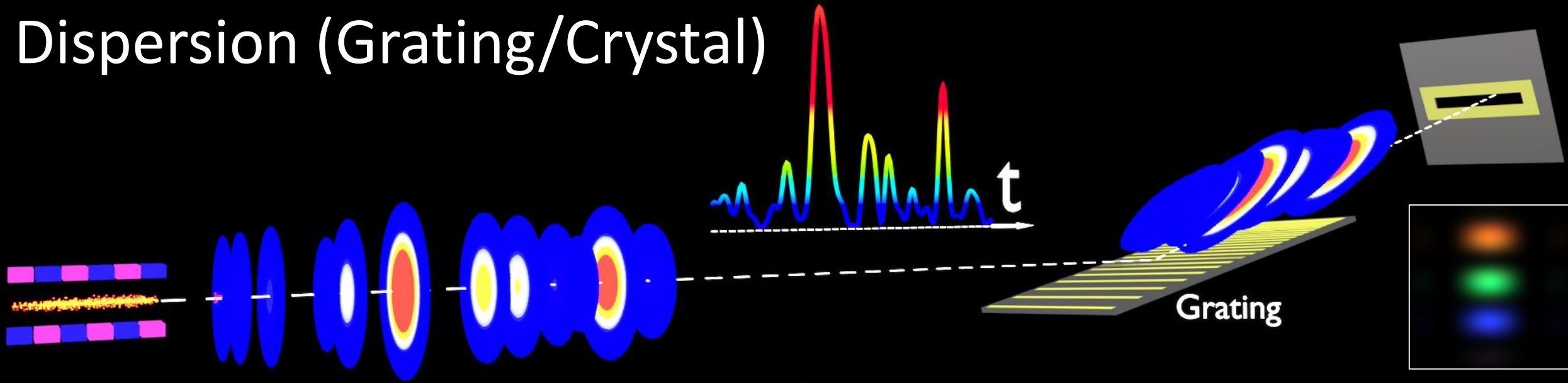
Σ'_S – pitch vibrations of the source

$$\zeta^{CF}(\omega) = \frac{\lambda/2\pi}{\Sigma_{ph} \cdot \Sigma'_{ph}}$$

Vibration – Source and Optics

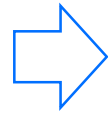


Dispersion (Grating/Crystal)

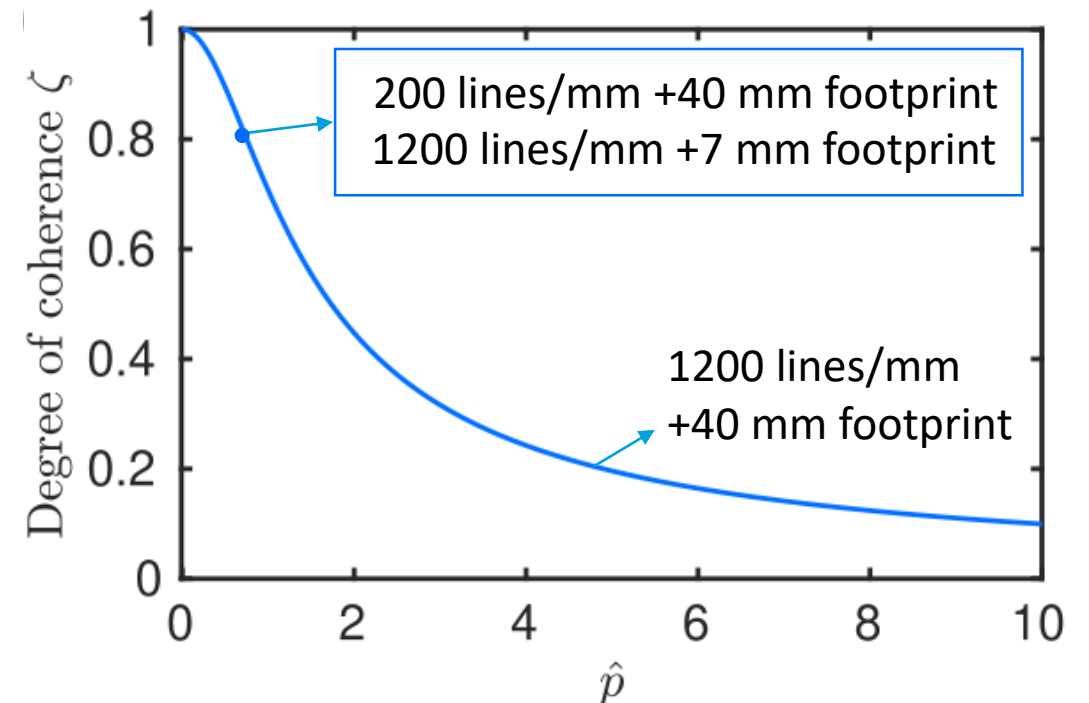


Degree of coherence

$$\zeta_g = \frac{1}{\sqrt{1 + 2p^2\sigma_\omega^2\sigma_x^2}} = \frac{1}{\sqrt{1 + \hat{p}^2}}$$

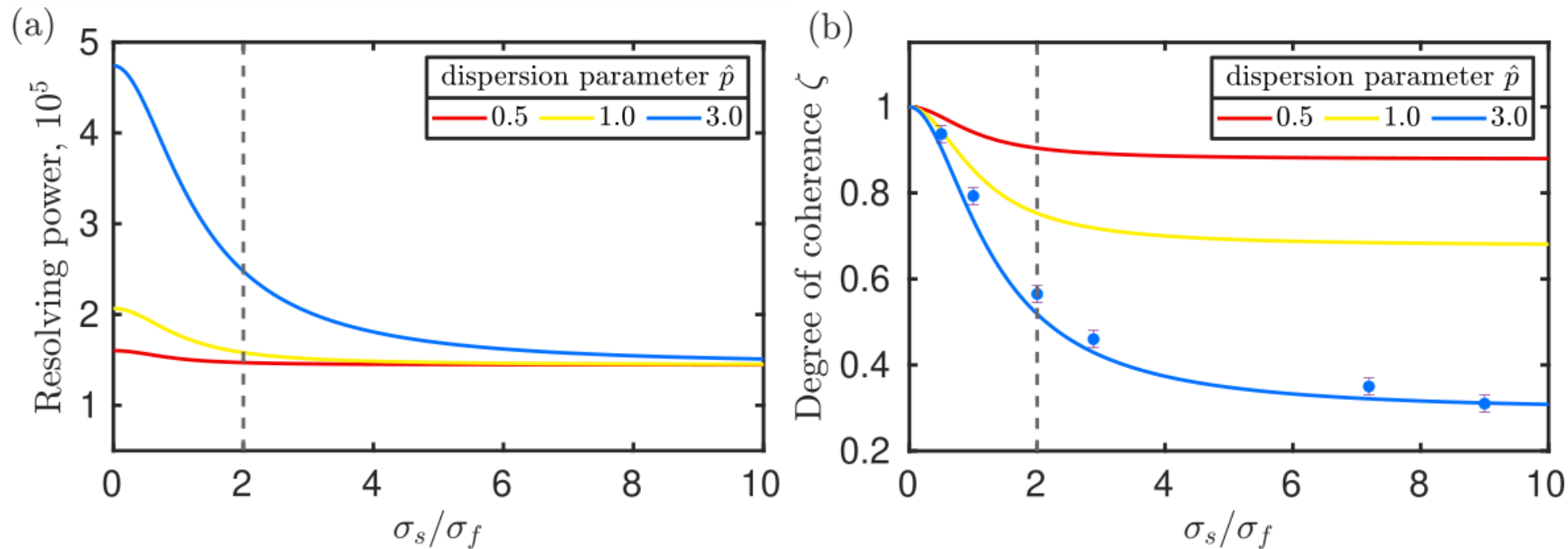


- The higher the dispersion parameter $\hat{p}$ of the grating, the lower the spatial coherence of the diffracted field.
- With strong dispersion, a spatial coherence degradation of ~80% can be achieved.



Dispersion (Grating/Crystal)

Resolving power and Coherence in focus with ES



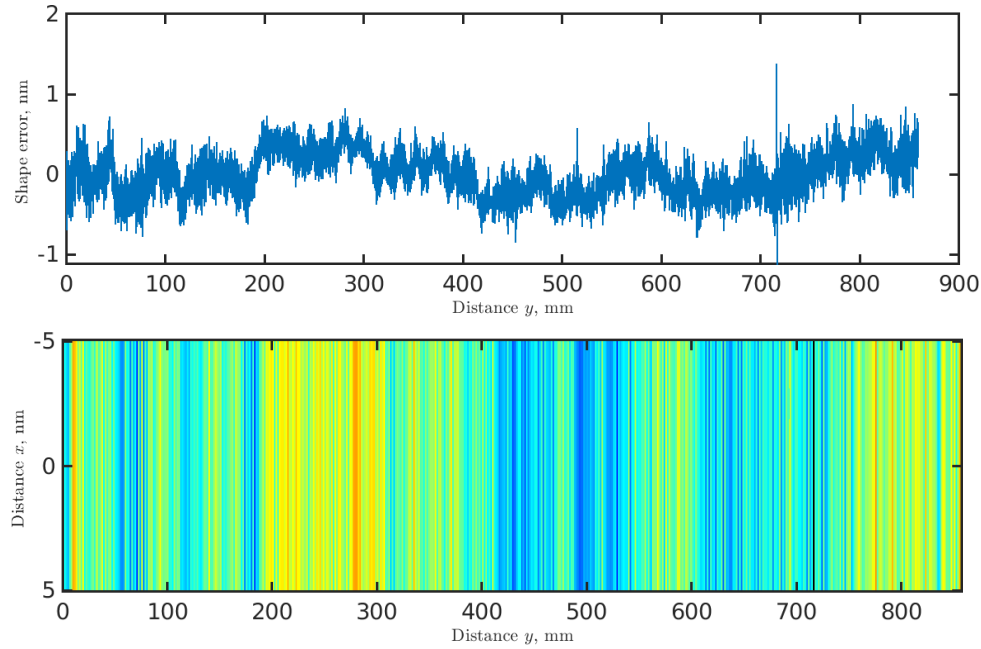
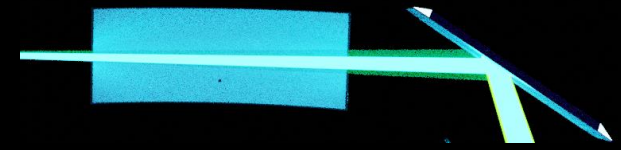
Degree of coherence

$$\zeta_{es} = \sqrt{\frac{\sigma_s^2 / (\sigma_f^2 \tilde{p}) + 1}{\sigma_s^2 / \sigma_f^2 + 1}}$$

Transmitted bandwidth:
$$\sigma_{\omega_s} = \frac{1}{\sqrt{2\sigma_x^2 p^2}} \left[\left(1 + \frac{\sigma_s^2}{\sigma_f^2}\right)^2 + \frac{\sigma_s^4}{\sigma_f^2 \sigma_x^2} \right]^{\frac{1}{2}} \left[1 + \frac{\sigma_s^2}{\sigma_f^2} \left(1 + \frac{\sigma_s^2}{\sigma_x^2}\right) \right]^{-\frac{1}{2}}$$

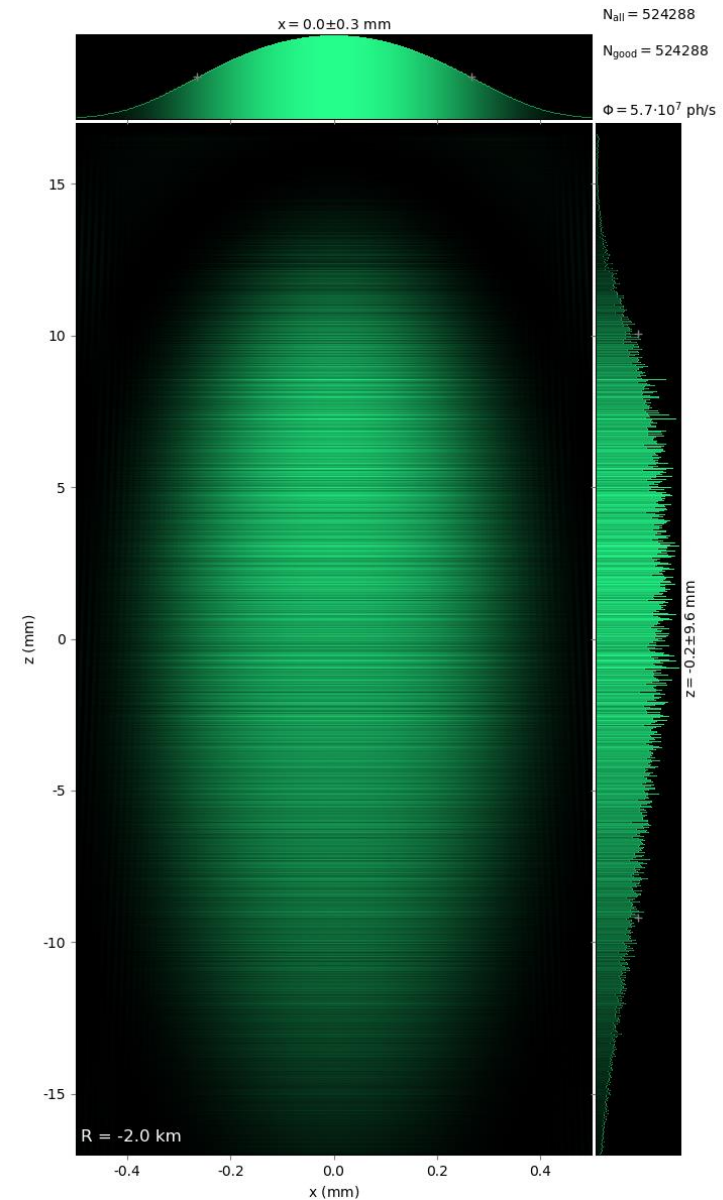
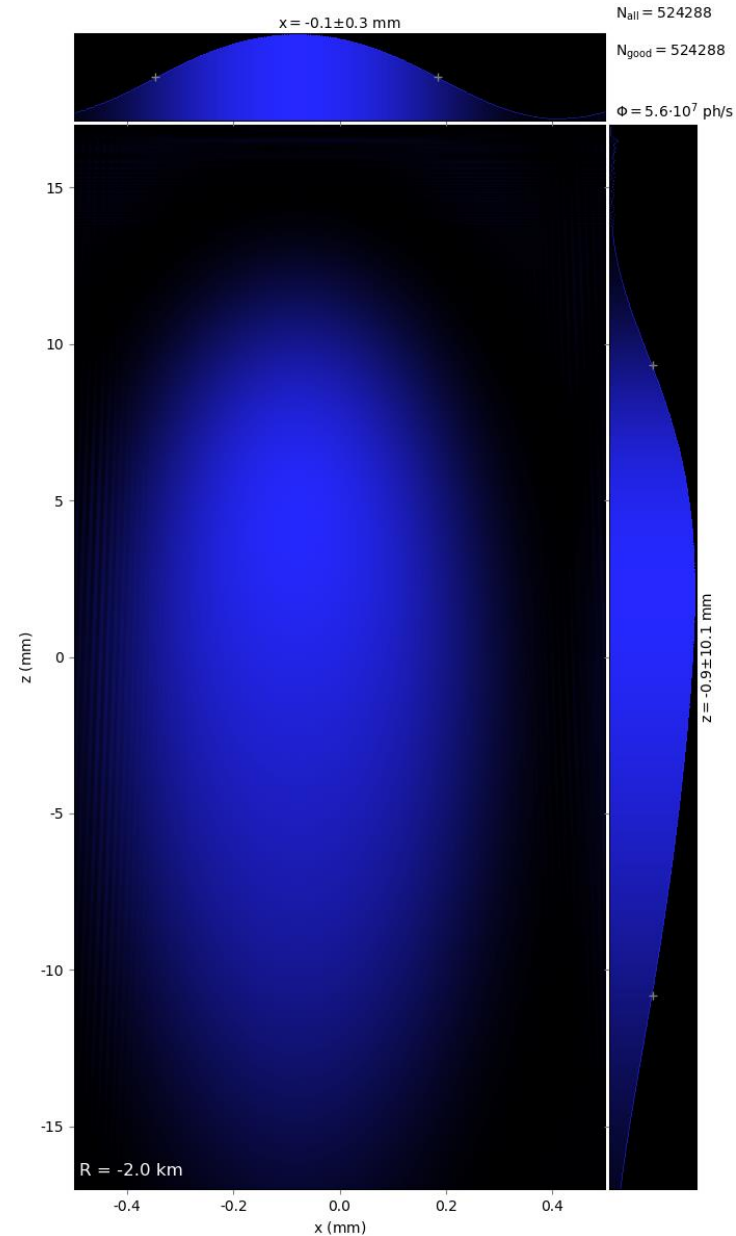
- For a given exit slit aperture opening, the degree of spatial coherence is **inversely** proportional to the resolving power of the monochromator.

Mirror surface

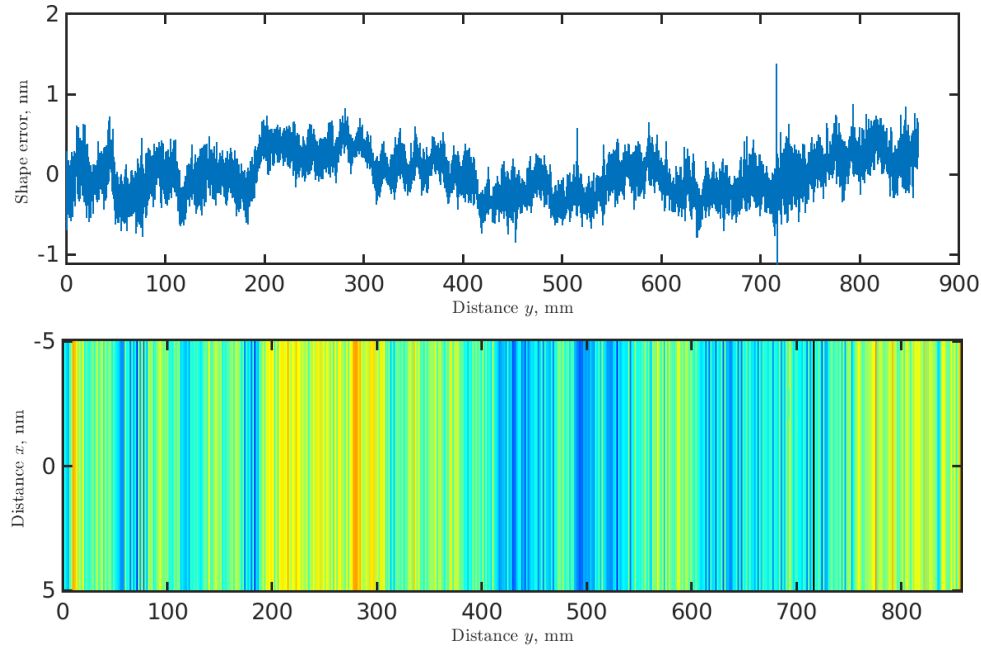
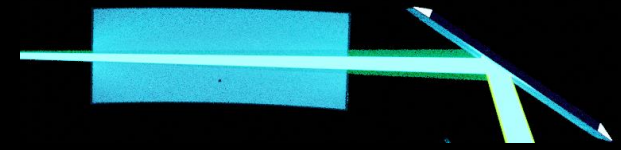


Energy: 50keV ($n=11^{\text{th}}$), $p=60$ m, $q=50$ m,
Emittance = 20×4 pmrad, Beta = 2m, $N_e = 100$

- The beam profile is disturbed (static case)

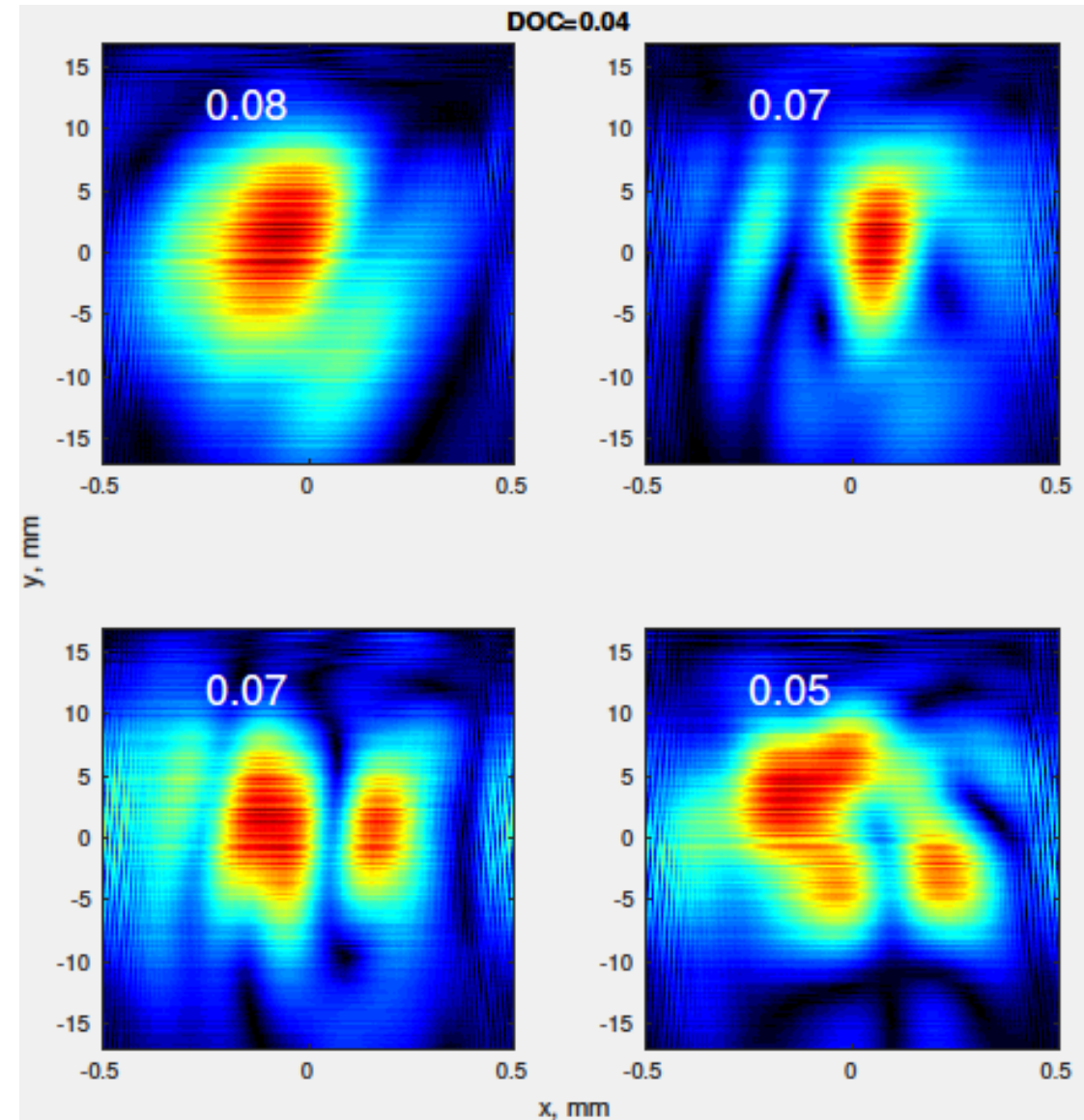


Mirror surface

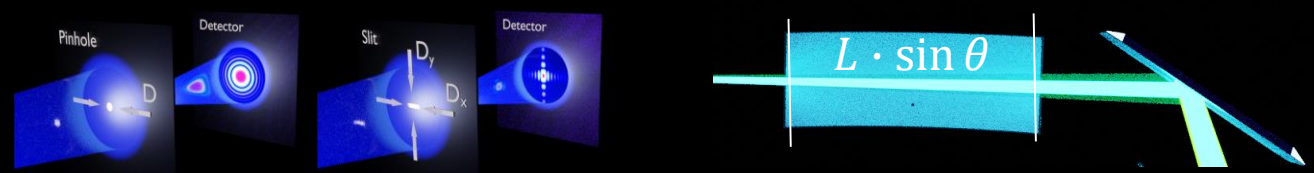


Energy: 50keV ($n=11^{\text{th}}$), $p=60$ m, $q=50$ m,
Emittance = 20×4 pmrad, Beta = 2m, $N_e = 100$

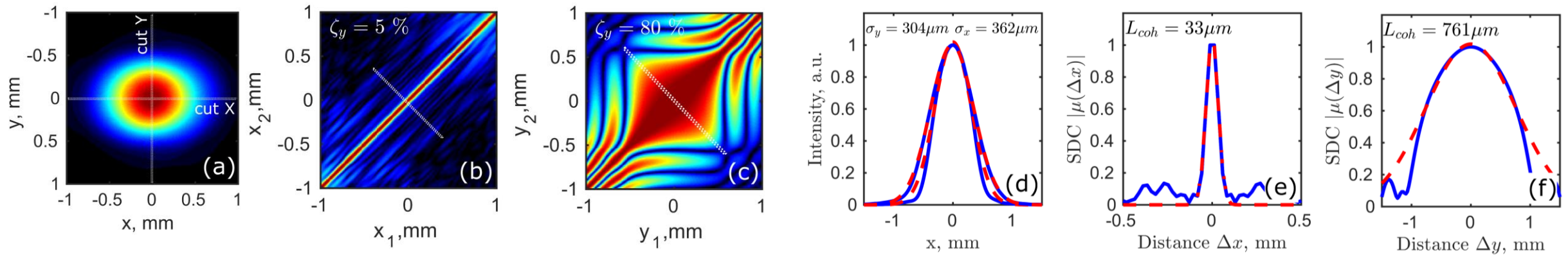
- No significant degradation of coherence due to surface errors (static case)



Aperture



The Degree of Coherence, Coherent Fraction, and the Coherence length of the source or at a distance can be determined by various methods:



Degree of coherence:

$$\zeta^{DC}(\omega) = \frac{\sum \beta_j^2(\omega)}{(\sum \beta_j(\omega))^2}$$

$$\zeta^{DC} = \frac{\iint |W(r_1, r_2)|^2 dr_1 dr_2}{\int I(r_1) dr_1 \int I(r_2) dr_2}$$

$$\zeta_i^{DC} = 1 / \sqrt{1 + \left(\frac{2\Sigma}{L_c} \right)^2}$$

Coherent fraction:

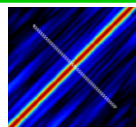
$$\zeta^{CF}(\omega) = \frac{\beta_0(\omega)}{\sum_{j=0}^{\infty} \beta_j(\omega)}$$

$$\zeta^{CF} \approx \frac{2\zeta^{DC}}{1 + \zeta^{DC}}$$

$$\zeta^{CF}(\omega) = \frac{\lambda/2\pi}{\Sigma_{ph} \cdot \Sigma'_{ph}}$$

Coherence length:

From SDC



$$L_c \approx \frac{\lambda z}{2\pi d}$$

$$L_c = 2\Sigma\zeta^{DC} / \sqrt{1 - \zeta^{DC^2}}$$

Transverse coherence length

- Another way of defining L_c comes from the phase space area. For a Gaussian laser mode with rms size and angle width of σ and σ' , phase space area (fully coherent)

$$\sigma\sigma' = \lambda/4\pi$$

- Considering a rectangle of width (coherence length) $L_c = \sqrt{2\pi}\sigma$ and height 1 has the same area as a Gaussian of rms width σ and height 1, we have

$$L_c\sigma' = \lambda/(2\sqrt{2\pi})$$

- Note: σ' is the angle coming to a single point at the position of interest. If the angular profile is not Gaussian but rectangular shape with width Δ' , similarly $\Delta' = \sqrt{2\pi}\sigma'$, then

$$L_c\Delta' = \lambda/2$$

- Coherence by propagation (van Cittert-Zernike theorem) of an incoherence source with Gaussian distribution profile of rms size Σ .

$$L_c\sigma' = L_c \frac{\Sigma}{D} = \frac{\lambda}{2\sqrt{2\pi}} \Rightarrow L_c = \frac{\lambda D}{2\sqrt{2\pi}\Sigma} \approx 0.2 \frac{\lambda D}{\Sigma}$$

